Sensitivity Is Not Fragility: Shared Visual Responses and Task Outcomes in Vision-Language-Action Policies Supplementary Material

Anonymous CVPR submission

Reading Guide and Measurement Definitions

This guide maps each main-paper claim (C1–C4) to its evidence and key numbers (Part 1), each stated limit to its documentation (Part 2), and indexes the studies and terms (Part 3). Recording sites and interval conventions: Sec. S1; protocol and recomputation checks: Sec. S15. Results are for $\pi_{0.5}$ unless stated.

1. Evidence for each main claim. C1: perturbations change the features along 16 shared directions per layer (main Sec. 4.1, Fig. 3a).

Sec. S3, Table S5. Held-out tasks and families, all 536 pairs: 36.7–84.2% of the energy around the fitting-data mean, against 0.5–10.6% (clean-feature) and 1.3–1.9% (random); 16.26–75.00% of the raw energy.

Sec. S3.4. The 404-pair subset of main Fig. 3: 36.5–83.6%, against 0.5–10.2% and 1.3–1.9%.

Table S4. Not the mean: uncentered shared directions retain 32.4–75.3% of the raw energy, the span of the position means 13.5–30.4%.

Sec. S3.3, Figs. S1, S2. Same ordering for 1 to 32 directions.

Secs. S17, S14, S8.1. Energy-weighted tasks: 38.4–84.0%. π_0: 39–77%, against 1–28% and about 1.5%. A later observation (original LIBERO; tasks held out, conditions seen in fitting): 32.7–64.2%, against 0.8–11.0% and 1.4–1.8%.

Sec. S2. Each reconstruction uses 160 coefficients from the response it scores.

C1, linear readout (main Fig. 3b).

Sec. S3.4, Table S6. The 16 coordinates predict the command change better than no change at L18 in all families, worse at L1 and L9 in 7 of 8 cells. At L18 they match clean-feature coordinates (intervals span zero); first flow predicts best in every row.

Sec. S13. Nonlinear readouts keep the layer ordering but change 102 and 76 of 216 interval categories.

C2: restoration at L9 reduces command error (main Sec. 4.2, Figs. 1b, 4).

Sec. S12, Tables S24–S26. 216 held-out cases: error 0.409 to 0.312 (Camera), 0.209 to 0.199 (Sensor Noise), lower in 25 of 27 and 24 of 30 tasks. Task success is higher by 1.85 and 3.33 points. The 16 directions reach 86.9% and 70.7% of replacing the whole L9 state, which gains 7.41 and 5.83 points (post hoc).

Sec. S12.5. At matched edit size, restoration beats clean-feature directions under Camera only.

Sec. S11. No clean observation (6,048 planned rollouts): L18 response lower by 8.8% and 7.6%; random directions at most 0.3%.

C3: command changes reach the robot (main Sec. 5.1).

Sec. S5.1, Fig. S8. Rank correlation with end-effector separation 0.95/0.97 after 5 controls, 0.47/0.41 after 60.

Sec. S7. Agreement across noise draws 0.88–0.92 and 0.80–0.89; lower for the gripper.

C4: the size of a change is not its cost (main Sec. 5.2, Fig. 5).

Sec. S5, Table S10. π_0, Sensor Noise: 0.58 times a clean rerun's motion, 14.3 points lost; every lost case listed.

Sec. S6. Lost and gained successes against the clean rerun; bounds for 39 of 2,680 missing outcomes. Lost successes exceed the rerun level under Camera and Sensor Noise, gained ones do not. AUROC 0.72/0.64 pooled, 0.62/0.55 within a family (post hoc).

Secs. S9.1, S9.2. Rank correlation -0.90 across ten condition means, -0.075 across 400 cells. Headlight: 5.1 times the command change of matched dimming, both 77 of 80 (Fig. S6).

Sec. S16. Complete main-paper panels.

Additional studies, not relied on by the main paper: internal associations (Sec. S4, Table S7), gripper timing (Sec. S7.1), LIBERO-Plus outcomes (Sec. S10).

2. Where the main paper's stated limits are documented (main Sec. 6).

Sensitivity is measured at the episode start: Sec. S1; the command–motion link weakens by 60 controls (Sec. S5.1); one later observation (Sec. S8.1).

Restoration: one layer, 16 directions, two families, 216 cases, needs a clean observation: Sec. S12; the test without one: Sec. S11.

The shared directions carry the Background response, which costs $\pi_{0.5}$ no net success: Table S5; Sec. S6.

Simulation, two related policies, moderate settings: Table S2; Sec. S10.

3. Study index.

Internal recordings ($\pi_{0.5}$, LIBERO-Plus): 536 pairs on 40 tasks fit the shared directions (Secs. S3, S10); 404 pairs on 28 common tasks link features, commands, motion and outcomes (Sec. S4).

Restoration ($\pi_{0.5}$): 31 held-out tasks, 108 settings, two draws, six runs, 1,296 rollouts (Sec. S12).
Paired executions ($\pi_{0.5}$, π_0): five runs per setting; 268 settings per policy on 40 tasks, 202 on 28 for outcomes (Sec. S6). Same settings as the recordings; π_0 internals regenerated on 369 of 404 pairs (Sec. S14).
Original LIBERO ($\pi_{0.5}$): later observation, 3,360 episodes (Sec. S8.1; its L9 directions are the ones removed in Sec. S11); removal without a clean observation, 288 cases $\times$ 3 conditions (clean, Camera $+15^\circ$, headlight off) $\times$ 7 runs; matched lighting, 80 initial states (Sec. S9.2); ten conditions, 2,344 of 2,400 pairs on 40 tasks.

Glossary.

Setting, pair, case. Setting: one task under one visual variant. Pair: one setting under one policy-noise draw (label 0 or 1); its clean and perturbed runs share task, start state and noise. Case: a pair in the restoration study. Start state: task, initial state and, where used, noise label.
Runs. Clean R_0, R_1; perturbed T_0, T_1 (noise of R_0, R_1); R'_0 repeats R_0; clean rerun: R_1 vs. R_0. Restoration: N_C clean, N_T perturbed unedited, R_{resp} restoration, R_{clean}, R_{rand} controls, R_{full} whole L9 state.
Directions. Sixteen channel directions fitted to responses (shared; "Response" in tables), to clean features ("Clean") or random ("Random"); position-centered fits subtract the ten position means of the fitting data.
Energy shares. One minus reconstruction error over the energy around the fitting-data mean (main Eq. 3), or over the raw energy.
Computation. Query: one observation and plan; generation step $j = 0, \ldots, 9$; plan position: one planned action; control: one executed action. First five commands: their translation part, 15 numbers. First flow: clean–perturbed difference of the step-0 update.
Internal quantities. Prefix H, normalized Z, cached values V; action-expert H after a block; attention probability P; attention output W before gating (Table S1). Conditional TV: redistribution within a region. Participation: effective number of channels (tokens) carrying a response. Raw output response: a difference RMS; relative response divides it by the matched-clean norm.
Command change, footprint. Perturbed minus clean first five commands; an edit's footprint is its edited-minus-unedited change as a fraction of it.
Outcomes. Clean then perturbed (S success, F failure): SF lost, FS gained success. Success change (RD in some tables): perturbed minus clean success rate. Command–cost correlation: across the ten conditions.
Intervals resample whole tasks. **Bounds over missing outcomes** cover all feasible values of unknown outcomes.
Difficulty level: LIBERO-Plus level 1–5, higher is harder. **Camera:** Camera Viewpoints family; on original LIBERO, $+15^\circ$ unless stated.
Abbreviations. RMS root mean square; MSE mean squared error; EEF end effector; TV total variation; AUROC (AUC in some tables) area under the ROC curve; pp percentage points.

S1. Policy Computation, Recorded Quantities, and Evaluation Design

This section states where each quantity is recorded (Table S1), which study compares it (Table S2) and with what sampling unit and intervals. For C1, the shared directions are fitted to the action-expert state H after L1, L9 and L18, at the first query and first generation step of the LIBERO-Plus recordings (main Figs. 1a and 2).

Each study asks a different question. Reconstruction asks how much of an observed response fitted directions retain (information, mean and capacity controls: Secs. S2 and S3); command readouts, what the response predicts about the commands (Sec. S3.4); associations, how quantities covary on matched cases (Secs. S10.2 and S4). Paired executions measure repeatability, motion and outcomes (Secs. S7, S5 and S6). The later-observation and original-LIBERO studies extend these questions to other states and schedules (Secs. S8 and S9).

Policies and evaluation design. The two policies generate plans of different length, 10×32 for $\pi_{0.5}$ and 50×32 for π_0, but both use ten generation steps and execute five controls before replanning. The paired executions reuse the 268 settings of the LIBERO-Plus recordings. Their assignments were fixed before their outcomes were read, but after the results of the LIBERO-Plus recordings were known.

Hidden states come from the policy's own run; attention comes from a replay. The action-expert states H are recorded during the policy's own generation, so they are what the policy computed. The attention quantities P and W are not stored in that pass; a diagnostic replay recomputes them from each branch's saved plan x_j and generation time, with the same frozen weights. This holds throughout the LIBERO-Plus analyses (Table S1). Section S15 compares the hidden states and velocities of the replay with the policy's own; no such comparison exists for P or W.

Sampling units and intervals by study. The shared-direction reconstruction is exploratory; it holds out both task and family and reports all 12 family/layer cells. Its unit is the task component: tasks that share a clean scene, held out together. Background, Camera, Light and Sensor Noise contribute 34/32/40/39 task components and 118/112/152/154 cases; Light includes 18 records without a LIBERO-Plus difficulty level. Its 95% intervals resample task components (2,000 draws), condition on the fitted directions and carry no multiplicity adjustment. Each query keeps 160 coefficients across ten positions.

The common-task comparison places internal summaries and behavior on the same 28 tasks and 404 pairs. Hidden-

Table S1. **Each internal quantity is read at one site through one recording path.** Definitions for measurements at the first query in the LIBERO-Plus recordings; widths are channel counts; P and W exist only in the diagnostic replay. The paired executions use the records in Sec. S6.

Quantity	Computation site	Recording support
Prefix H / Z	Pre-block state / pre-attention RMSNorm output; width 2048	Recomputed prefix; layers 1, 9, 18; image/wrist/text masks
Prefix V	Projected cache values; one KV head of width 256	Recomputed once per query; no generation-step axis
P	Full-key probability	Diagnostic suffix; eight heads, ten action queries; steps 0, 5, 9
W	Projected attention output before the adaptive gate	Diagnostic suffix; ten positions of width 1024; steps 0, 5, 9
Action-expert H	Post-block state of width 1024	Policy's own run; layers 1, 9, 18
Commands	First five commands returned after the output adapter	Fifteen translation coordinates; command units

Table S2. **Each study has its own sampling unit and support.** Rows are distinct studies of the frozen policies ($\pi_{0.5}$ unless stated); support gives planned and, where records are missing, available counts. Studies may share tasks or clean episodes but keep their own executions.

Study	Recorded support	Main role
Matched-lighting study	80 initial states; 240 episodes	Headlight versus matched dimming
Ten-condition study	2,344/2,400 persistent pairs; 40 tasks	Response–command association and aggregation effects
LIBERO-Plus recordings	536 pairs; 40 tasks; 4 families	Shared directions and readout (C1); matched internal, command, motion and outcome measurements on 404 pairs / 28 tasks
Restoration	31 held-out tasks; 108 Camera/Sensor Noise settings; 1,296 episodes	Restoration at L9 at every generation step (C2); first five commands and complete paired task outcomes
Paired executions	268 planned settings per policy; 536/536 policy–setting cases archived	Paired behavior of both policies (C3, C4); 2,641/2,680 outcomes valid

state and attention summaries are descriptive means. Commands and motion have pointwise 95% task-block intervals; outcomes keep the four-family Bonferroni adjustment for simultaneous 95% coverage. Motion ratios of task-equal means have no interval here; Sec. S10.3 gives the paired motion differences.

The paired executions estimate the paired success change against two clean runs on the prespecified primary support of 28 tasks and 202 settings per policy, with fixed noise weights. All 536 policy–setting cases are archived; 39 run outcomes are unknown (25 technical failures and 14 that fail the analysis validation criteria) and enter bounds over missing outcomes. The 2,000 task-bootstrap draws give four-family-adjusted 98.75% intervals for the success changes (Sec. S6).

Implementation notes. The policy's forward pass applies all 18 blocks through a rematerialized scan. The diagnostic replay uses a separate full-depth scan, retains blocks 0, 8 and 17 (L1, L9 and L18), and takes the saved visibility masks as inputs. Final actions of the policy's own run are checked against the reference baseline before H is adopted; diagnostic readouts and velocities are checked exactly against in-graph controls.

The replay shows how the allocation P becomes the projected output W. It recomputes the prefix; normalized suffix states give queries, keys and values, with rotary embeddings on queries and keys and head-dimension scaling of queries. Cached prefix keys and values are concatenated with the suffix's, and softmax spans all visible keys before value aggregation. W is the output projection before the adaptive attention gate and residual addition; the feed-forward sublayer and its gated residual then give post-block H. Prefix V stays fixed within a query; suffix V changes with the generation state. Masked attention logits are accumulated at higher precision and cast to the model precision before value aggregation.

S2. Information Used for Reconstruction, Prediction, and Readout

Reconstruction, the C1 test, observes the response it scores: the 16 directions are fixed in advance, but their 160 coefficients per query come from the test response. C1 therefore shows that fixed directions, fitted without the test tasks and

family, capture new responses, not that a response can be predicted unobserved. Table S3 lists templates and the command readout for contrast.

Reconstruction observes the response it scores. For one layer, $D \in \mathbb{R}^{10\times1024}$ is the paired response over ten plan positions, M a broadcast global training mean or a ten-position training template, and $B \in \mathbb{R}^{1024\times16}$ an orthonormal channel basis. Fixed-template prediction is $\widehat{D} = M$; reconstruction projects the residual of the observed response onto the basis:

$$\widehat{D} = M + [(D - M)B]B^{\top}. \tag{S1}$$

The 10×16 coefficients are computed from the test response. Sixteen directions therefore give 16 coefficients at each position, 160 per query, in addition to the stored directions and template. The test evaluates reuse of the directions, not of the coefficients.

The clean-feature control learns directions from training clean features but projects the same test residual around the same template, so both sets of directions have the same target and test information. Mean-span controls, projecting onto the span of training means, also observe the target through their projection coefficients, unlike mean-only prediction: a global mean direction uses ten coefficients, a rank-ten span of the position means 100.

Command readout compares response coordinates with the first flow output. The first-flow baseline is the clean–perturbed difference Δv_0 in the action expert's first update (initial predicted velocity), with 320 coordinates. Response coefficients contribute 160 coordinates; their concatenation with Δv_0 contributes 480. Comparing the concatenation with first flow alone tests the additional command information available to the fitted readout. Only sets of directions of equal size match the coefficient budget. Final commands, measured motion and task success remain separate endpoints.

S3. Shared Directions Beyond Training Means

This section is the main evidence for C1 (main Sec. 4.1). On held-out tasks and families of all 536 $\pi_{0.5}$ pairs, 16 shared directions retain 36.7–84.2% of the response energy around the fitting-data mean, against 0.5–10.6% for clean-feature and 1.3–1.9% for random directions (Table S5), for 1 to 32 directions alike (Sec. S3.3). The result does not come from the mean: uncentered shared directions retain more raw energy than training-mean templates and their spans (Table S4). Of the raw energy (no template subtracted) the centered fit retains 16.26–75.00%, less where the fitting-data mean misses the held-out responses. Section S3.4 gives the 404-pair subset quoted in the main paper.

S3.1. Shared directions against mean and capacity controls

Each fit holds out the tested tasks and family. The analysis uses the first query, first generation step, ten ordered plan positions and 1,024 channels at each of L1, L9 and L18, fitted separately. Tasks sharing a clean scene (a task component) are held out together, so no test scene is seen in fitting. The original 60 joint task/family splits and 12 known-task, held-family splits are kept. Training weights are equal at each nested level (task component, family, setting, noise, position); test folds are joined before a family result, so each held-out component appears once. The families contribute 118/112/152/154 pairs and 34/32/40/39 task components; the full Light set includes 18 cases without a LIBERO-Plus difficulty level. Inputs, numbers of directions and fits do not use outcomes.

Every set of directions reconstructs the same residual about a training template. A global mean μ and ten position means μ_p (the templates M) use training data only, broadcast across positions or applied per position. Centered shared directions are fitted to $D - M$; centered clean-feature directions to clean features after subtracting their own training clean means; both, and the random directions, reconstruct the same residual $D - M$. Mean-only prediction has no test coefficients; the mean-direction and position-mean-span controls project the observed target (Sec. S2), the latter with rank ten in every fold.

Scores are shares of energy retained. Higher is better; zero is no better than predicting zero (raw) or the template. For each held-out task component t, the score is one minus the reconstruction error SSE_t over the response energy, raw (E_t) or around the training template ($E_{M,t}$), averaged over components:

$$S_{\mathrm{raw}} = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}}\left(1 - \frac{\mathrm{SSE}_t}{E_t}\right), \tag{S2}$$

$$S_M = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}}\left(1 - \frac{\mathrm{SSE}_t}{E_{M,t}}\right). \tag{S3}$$

Global and position templates give different residual energies. Zero denominators and too few directions are undefined. All primary comparisons have valid inputs and enough directions.

Statistics. The tables report the primary comparisons with 16 directions on the original 72 training splits. Uncertainty uses 2,000 whole-task-component bootstrap draws shared across methods and numbers of directions, conditional on the fitted directions. All 95% intervals are pointwise: paired difference intervals for the 16-direction contrasts, marginal

Table S3. **Only reconstruction scores what it observes.** Columns: what each test observes, its scored target and its question. M is a training mean or position template, B a learned channel basis (for example the shared directions). Reconstruction and its clean-feature and random controls project the same test response. Task holdouts concern analysis fits, not policy pretraining.

Comparison	Information used to produce a test output	Scored target	Question
Fixed template	Training response mean or position template; no test response	Unobserved response D	Can a population template transfer?
Response reconstruction	Observed D; coefficients DB or $(D-M)B$	The same observed D	Do learned channel directions represent new responses?
Command readout	Paired first-query response features, paired first-flow difference, or both	Paired difference of the commands (before or after the output adapter)	What command information remains beyond strong baselines?

intervals for the curves. Resampling preserves the dependence among family/layer comparisons, which share data and training support.

S3.2. What remains after centering

The shared directions do more than the mean. The global and position mean-only predictors each have negative raw scores in seven of twelve cells (Table S4), yet the span of the position means reconstructs 13.5–30.4% of the raw response energy. Uncentered shared directions reconstruct 32.4–75.3%, with paired advantages of 18.2–57.5 percentage points over that span and positive pointwise intervals in all twelve cells. The span has rank ten, the shared directions sixteen; equal-size comparisons follow.

Centered shared directions keep a wide margin over clean-feature and random directions. With the training position template removed, the shared directions retain 36.7–84.2% of the energy around the mean, against 0.5–10.6% for centered clean-feature and 1.3–1.9% for random directions (Table S5). Around the global template they retain 35.8–84.3%. Every paired pointwise interval favors the shared directions under both raw and around-mean scoring. Projection reduces or preserves error relative to mean-only prediction.

Why raw and around-mean shares differ. With the position template, the shared directions retain 36.73–84.24% of the energy around the mean and 16.26–75.00% of the raw energy on this full 40-task panel (Table S5). The main paper quotes the first on the 404-pair subset (36.5–83.6%, Sec. S3.4) and the second from this table. Both divide the same reconstruction error: by the energy left after subtracting the training template, or by the raw response energy. They are close where the template, learned without the test family, matches the held-out responses (Sensor Noise; Camera at L9 and L18). Where it does not, subtracting it adds energy: mean-only prediction has negative raw scores (Table S4), so the energy around the mean exceeds the raw energy and the raw share is lower (Background, Light, Camera at L1; largest for Light L1). Centering removes only the training template; an offset shared within the held-out family stays in the scored response.

The effect of position centering depends on the family and layer. On the common raw denominator, position-minus-global reconstruction has negative intervals in six cells, a small positive interval in one, and intervals spanning zero in five. The positive cell is S/L9, $+0.151$ percentage points $[0.020, 0.281]$. For Background at L9 and L18, the position-centered raw shares, 20.91% $[-4.45, 35.50]$ and 16.26% $[-17.46, 34.83]$, are not distinguishable from zero.

Section S3.4 reads commands from the position-centered directions, fitted with the same splits.

S3.3. Other numbers of directions

The shared directions retain more than clean-feature and random directions for every number tested, $k \in \{1, 4, 8, 16, 32\}$ (Figs. S1 and S2), supporting C1 for other numbers of directions. It uses the 536-pair panel and all 72 training splits (60 joint task/family, 12 known-task, held-family). The 404 cases of Sec. S3.4 give the main paper's paired reconstruction–readout comparison.

Matched targets and nested directions. For every k, all three sets reconstruct the same residual $D_p - \mu_p$ about the training position mean, with the same denominator. The shared and clean-feature curves use the leading k directions of their training decompositions. Random directions are the first k columns of one fixed 32-column orthonormal basis, whose first 16 also give Table S5; the uncentered replication uses its own fixed random basis.

The ordering holds for all five numbers of directions. Shared directions retain more energy around the mean than clean-feature and random directions of equal number in every family/layer cell under both holdout settings. Under joint task/family holdout, the shared directions retain 7.0–66.0% with one direction and 49.3–88.9% with 32; with 32, clean-feature and random directions retain 1.0–15.0% and 2.8–

Table S4. **Uncentered shared directions retain more raw energy than training-mean templates and their spans.** Raw-energy scores S_{raw} (%) under joint task/family holdout in the LIBERO-Plus recordings ($\pi_{0.5}$), by family (B/C/L/S: Background Textures, Camera Viewpoints, Light Conditions, Sensor Noise incl. blur and fog-like conditions) and layer. Shared (PCA16): 16 uncentered shared directions. Mean-only columns use no test response; the others project it, with 160, ten and 100 coefficients. Negative scores mean greater error than predicting zero. Point values; pointwise 95% task-component intervals are in the text.

Family/layer	Shared (PCA16)	Global mean only	Position mean only	Global mean span	Position-mean span
B / L1	55.63	-47.41	-48.76	11.68	29.46
B / L9	33.86	-90.08	-112.16	2.52	14.60
B / L18	32.51	-94.23	-112.92	3.56	13.55
C / L1	62.33	-80.70	-81.72	5.78	28.96
C / L9	64.66	0.92	1.64	6.42	22.36
C / L18	42.96	4.20	3.28	9.05	20.64
L / L1	49.27	-272.46	-277.68	3.06	30.43
L / L9	37.86	-68.91	-81.52	3.30	14.84
L / L18	32.37	-47.67	-56.72	3.53	13.63
S / L1	75.27	4.00	4.03	11.03	17.75
S / L9	47.16	4.45	4.80	12.07	24.82
S / L18	38.59	3.48	1.69	11.34	20.37

Table S5. **Shared directions retain far more than clean-feature and random directions after centering; raw and around-mean shares.** Percent; same panel and holdouts. First two columns: raw shares S_{raw} of the 16 shared directions after global or position centering. Other columns: shares S_M around the global or position training template for 16 shared (Response), clean-feature (Clean) and random (Random) directions, each with 160 coefficients from the test response. Means of task-component ratios; pointwise 95% intervals are in the text. Both shares have the same error and different denominators: Light L1 retains 84.24% around the position template and 45.48% raw.

	Raw share, shared directions		Share around global template			Share around position template		
Family/layer	Global	Position	Response	Clean	Random	Response	Clean	Random
B / L1	54.87	54.63	65.19	2.53	1.33	65.28	2.53	1.33
B / L9	24.95	20.91	43.18	0.57	1.45	44.39	0.53	1.50
B / L18	24.65	16.26	38.55	9.65	1.60	39.30	8.12	1.63
C / L1	60.17	59.96	77.07	1.73	1.39	77.08	1.75	1.39
C / L9	64.92	64.82	64.07	0.53	1.65	63.70	0.50	1.63
C / L18	42.73	42.49	40.14	13.22	1.70	40.43	10.55	1.72
L / L1	46.42	45.48	84.28	1.91	1.85	84.24	1.90	1.84
L / L9	35.05	31.03	60.64	0.85	1.57	60.87	0.82	1.67
L / L18	29.16	24.19	50.85	7.72	1.92	50.55	6.90	1.90
S / L1	75.00	75.00	73.97	1.35	1.62	73.98	1.35	1.62
S / L9	46.81	46.96	43.73	0.61	1.39	43.46	0.61	1.39
S / L18	38.11	37.89	35.84	9.60	1.86	36.73	8.73	1.88

3.8%. Paired intervals against the controls are reported for the primary $k = 16$, where every interval favors the shared directions in both settings.

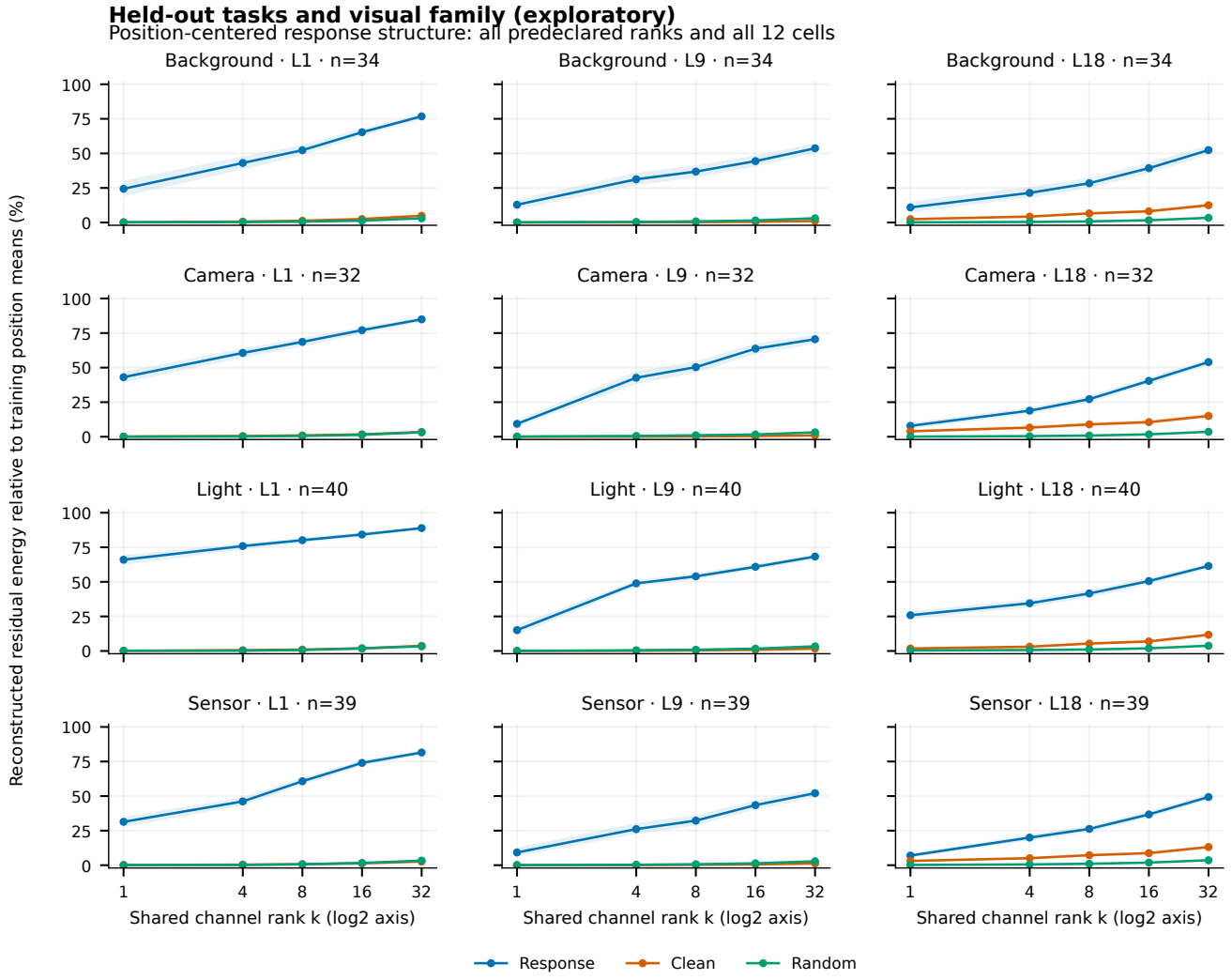


Figure S1. **Shared directions retain more than clean-feature and random directions for every number of directions, under joint task/family holdout.** Shares of the energy around the training position means on the original 536-pair LIBERO-Plus panel, for $k = 1, 4, 8, 16, 32$ directions. Twelve panels: four families by three layers; $34/32/40/39$ task components for Background/Camera/Light/Sensor Noise. Curves average task-component scores. Shading: pointwise 95% intervals from 2,000 bootstrap draws of whole task components, conditional on the fits. Each set uses k coefficients of the observed response per position, $10k$ in total.

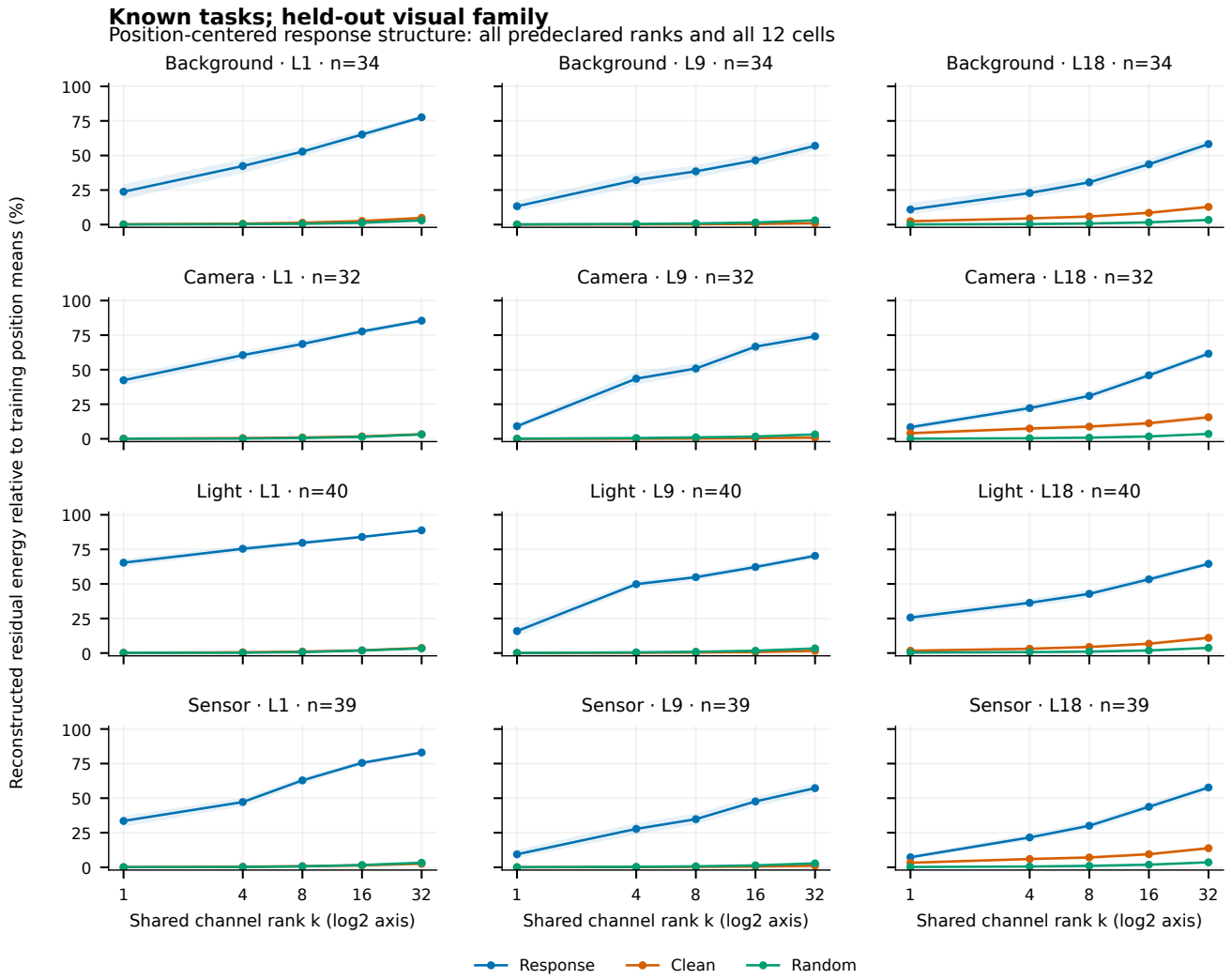


Figure S2. **The same ordering holds for known tasks and a held-out family.** The 536-pair panel under the twelve known-task, held-family splits; same target, denominator and $34/32/40/39$ task components as Fig. S1. Shading: pointwise 95% intervals as in Fig. S1, using the same 2,000 draws. Each set uses k coefficients of the observed response per position, $10k$ in total.

S3.4. The 404-pair subset and the linear command readout

On the 404 pairs of main Fig. 3, the position-centered shared directions retain 36.5–83.6% of the response energy around the fitting-data mean, against 0.5–10.2% for clean-feature and 1.3–1.9% for random directions: the C1 numbers of the main paper. An exploratory linear readout of the same 16 coordinates predicts the change in the first five commands better than predicting no change at L18 in all four families, but not at L1 and L9 (Table S6; main Fig. 3b). This supports main Sec. 4.1 with two qualifications: at L18 the shared coordinates predict no better than clean-feature coordinates, and first flow predicts better than any set of coordinates.

What the readout is compared with. Affine readouts are compared on the same held-out cases with predicting no change (absolute accuracy), clean-feature and random coordinates (choice of directions), the full hidden response (prediction before compression) and first flow (information beyond a downstream output; it beats no change in every family). Table S6 puts all layers on one translation-error scale; Table S29 adds rotation and gripper.

At L1 and L9, better than clean-feature coordinates, worse than no change. At L1 and L9, shared-direction coordinates reduce affine translation error by 20–75% and 27–71% relative to centered clean-feature coordinates. On the absolute scale, however, the shared-direction readout has higher error than predicting no change in 7 of these 8 cells, and the clean-feature readout in all 8 (point estimates). The first compares coordinates within a layer; the second asks whether a readout beats keeping the clean commands. Against the full-hidden readout, 8 of 12 translation intervals include zero (3 favor the shared directions, 1 the full readout).

Within most tasks, L1 retains most and L18 predicts best. We compare retained response energy and affine translation-prediction MSE at L1, L9 and L18 within each task, on the same pairs. In the table below, *best-layer separation* counts tasks in which L1 retains strictly more response energy than both other layers and L18 has strictly lower prediction MSE than both. This holds in 24–27 of the 28 tasks in each family. The *full ordering* column also requires both retention and prediction MSE to decrease strictly from L1 through L9 to L18, and so also checks the position of L9; it holds in fewer tasks. The same 28 tasks appear in all four families. Counts are descriptive, without an inferential test.

Family	Best-layer separation	Full ordering
Background	24/28	11/28
Camera	26/28	18/28
Light	27/28	15/28
Sensor Noise	27/28	17/28

At L9 the advantage over clean-feature coordinates depends on centering. At L9, shared-direction coordinates improve translation readout relative to centered clean-feature coordinates in each family. Uncentered clean-feature coordinates, however, give lower prediction error than the centered ones (Table S6), while the intervals for position-centered versus uncentered shared directions span zero in every family. The relative gain at L9 thus depends on how the clean-feature coordinates are built.

At L18 the shared directions do not beat clean-feature coordinates. At L18, translation and rotation readout differences against centered clean-feature coordinates have intervals spanning zero in every family. Continuous-gripper readout errors exceed those of the magnitude baseline (positionwise magnitudes) in every family. The pattern thus depends on the command component (contrasts against clean-feature coordinates: Table S29): reconstruction is scored once, prediction per component.

Adding shared coordinates to first flow rarely helps. Of the 36 first-flow-plus-shared-direction comparisons, one pointwise interval is positive, fifteen are negative and twenty span zero. The positive Camera/L1 translation contrast reduces error by 8.66%; the uncentered shared directions already give an 8.81% point reduction on the same support. Since first flow is already a downstream prediction, this measures additional command information available to the readout.

The directions are reused without refitting. The saved position-centered shared, clean-feature and random directions are used without refitting or choosing their number from command results; each has sixteen channel directions with coefficients at ten positions. The cached first-five adapter differences match all 35 returned command coordinates in each of the 404 cases. Translation, rotation and continuous gripper are scored separately; the gripper target precedes sign formatting. Comparators are the uncentered shared, clean-feature and random directions, ten positionwise response magnitudes, the full 10×1024 hidden difference, the first flow difference of the policy's own generation, zero prediction and the training target mean. The sixty joint task/family splits are kept, five held-out folds joined per family and layer. Training uses the LIBERO-Plus recordings; evaluation uses the matched subset on 28 tasks, with 96/102/96/110 cases for Background/Camera/Light/Sensor Noise.

Training the command readout. A weighted affine ridge readout predicts the target vector before or after the output adapter. Each feature block is centered on training data and divided by its training scalar RMS; the intercept is unpenalized and $\lambda = 0.01$ is fixed. Features are never pooled across layers or plan positions; adding the shared coordinates to first flow uses one training scale per block. The full-hidden readout has more features, so this compares prediction under fixed estimator and feature budgets.

Statistics. Command and reconstruction comparisons share the 404 test cases; within tasks, noise and setting weights follow the original protocol. Reconstruction divides task-weighted error by task-weighted residual energy about the saved training position mean and averages task scores. Readout contrasts compare task MSEs on the same 2,000 whole-task-component draws per family/layer and report the relative error reduction $1 - \mathrm{MSE}_{\mathrm{candidate}}/\mathrm{MSE}_{\mathrm{baseline}}$. Intervals are pointwise 95%, conditional on the fitted directions and readouts; the repeated family/layer/component comparisons share data.

Table S6. **The linear readout beats predicting no change only at L18, and there it matches clean-feature coordinates.** Task-equal translation MSE of the first five commands $\times 10^3$, in squared adapter-command units; lower is better. All columns use the same 404 held-out cases on 28 tasks. Response, Clean and Random are 16 position-centered shared, clean-feature and random directions; the last two columns use uncentered fits. Full H uses the complete hidden response, and First flow the clean–perturbed difference in the initial predicted velocity. First flow and No-change are the same at every layer. Main Fig. 3b shows the errors of the 16-direction sets relative to No-change; First flow is kept here as a downstream comparator. Paired readout intervals appear in Table S29 and Fig. S3.

Layer	Family	Response	Clean	Random	Full H	First flow	No-change	Uncentered response	Uncentered clean
L1	Background	4.297	10.213	7.298	6.877	0.453	2.084	4.282	9.030
L1	Camera	16.280	20.428	18.038	15.072	1.258	15.414	16.263	19.629
L1	Light	1.875	7.400	3.937	3.578	0.227	1.063	1.881	6.898
L1	Sensor Noise	18.564	27.219	29.106	30.301	0.655	4.609	18.978	20.971
L9	Background	3.825	7.597	5.069	2.771	0.453	2.084	3.935	3.423
L9	Camera	14.407	19.831	16.383	13.418	1.258	15.414	14.442	17.308
L9	Light	1.365	4.713	2.478	1.693	0.227	1.063	1.351	2.278
L9	Sensor Noise	6.331	14.484	8.915	6.484	0.655	4.609	6.193	7.433
L18	Background	1.035	0.964	4.194	0.686	0.453	2.084	1.040	0.872
L18	Camera	3.737	3.457	11.073	4.349	1.258	15.414	3.940	3.568
L18	Light	0.433	0.496	1.482	0.389	0.227	1.063	0.416	0.435
L18	Sensor Noise	1.538	1.480	4.028	1.510	0.655	4.609	1.564	1.268

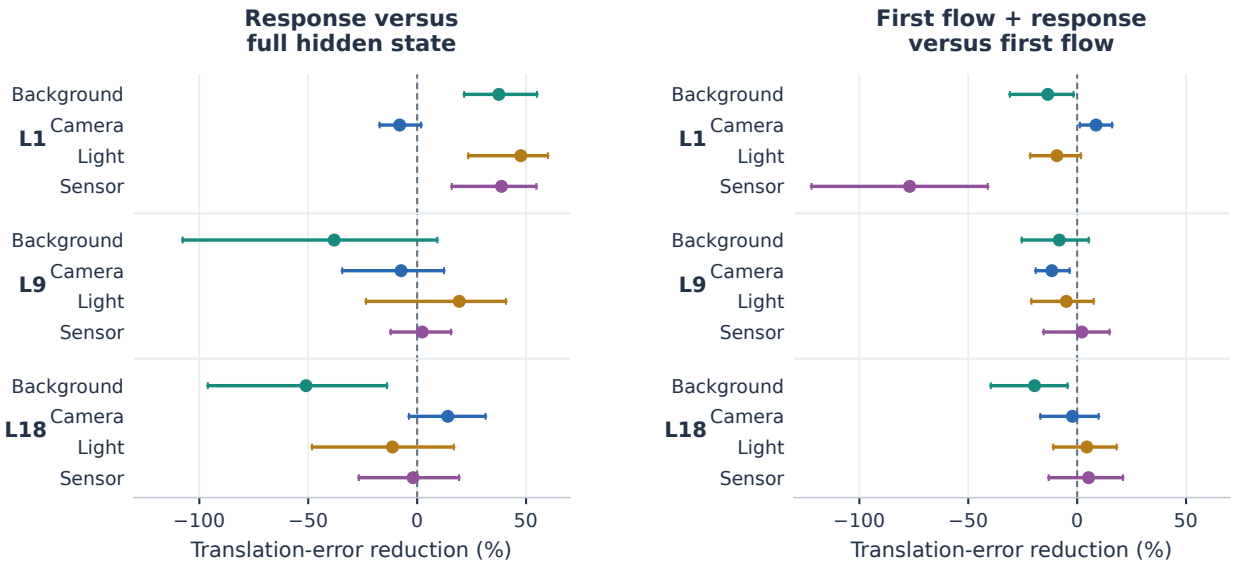


Figure S3. **Shared coordinates mostly match the full hidden response and rarely add to first flow.** All 24 translation contrasts across three layers and four families, on the 404 matched LIBERO-Plus pairs and original task/family holdouts. Left: shared-direction coordinates versus the full hidden response. Right: first flow plus shared coordinates versus first flow. Error reduction is $100(1 - \mathrm{MSE}_{\mathrm{candidate}}/\mathrm{MSE}_{\mathrm{baseline}})$; positive favors the candidate. Bars are pointwise 95% task-component intervals under fixed fitted readouts.

S4. Associations between Internal Responses, Commands, and Outcomes

Do larger internal responses in $\pi_{0.5}$ go with larger command changes and with lost success? Under Camera and Sensor Noise, the L18 attention and action-expert responses rank with the change in the first five commands (Table S7). Their links to outcomes are weak, except that the attention output under Camera separates lost from kept successes. This is an additional, post hoc study; the main paper does not rely on it.

Pairs. The analysis uses 404 of the 536 planned $\pi_{0.5}$ pairs of the internal recordings on LIBERO-Plus: the 202 settings on the 28 common tasks, each with two policy-noise draws. All 404 have the four L18 signals, verified first five commands (translation part), motion after controls 5 and 60, and a paired outcome; the other 132 pairs are not used. Pairs are matched on case, task, setting, noise draw, clean run and outcome before any signal is selected, so the sample cannot depend on the signals. Section S10.3 reports motion on these pairs; the paired executions of Sec. S6 are separate episodes.

Signals and endpoints. The four signals are fixed, all at L18 and the first query: the block-input prefix state for the main camera (prefix H); the total-variation change of the attention probabilities over all keys (P); the attention output after projection and before gating (W), both recomputed; and the action-expert state from the policy's own generation (AE). P, W and AE are taken at the first generation step. Wrist and text prefix states and the cached values V (main camera, wrist, text; no generation-step axis) are companion signals (all recording sites: Table S1). The command endpoint is the RMS of the fifteen translation differences of the first five commands whose environment call returned, in command units. The signed outcome is $dY = Y_T - Y_R$: a lost success (SF) counts -1, a gained success (FS) $+1$.

Rank correlation within tasks. Ranks are centered within tasks, so that differences between tasks cannot create a correlation. Within a task with n_t pairs, each variable becomes $u_i = (\mathrm{rank}_{\mathrm{avg}}(x_i) - \frac{1}{2})/n_t$ (average ranks for ties), and pair i has weight $w_i = 1/(v_t n_{v(i)})$, where v_t counts the task's settings and $n_{v(i)}$ the observed noise draws in pair i's setting. After subtracting weighted task means, the task covariance is $C_t = \sum_i w_i \widetilde{u}_i \widetilde{z}_i$ and the variances $A_t = \sum_i w_i \widetilde{u}_i^2$, $B_t = \sum_i w_i \widetilde{z}_i^2$; the pooled coefficient is $\overline{C}/\sqrt{\overline{A}\,\overline{B}}$, averaging tasks equally. A variable constant within a task contributes zero variance and covariance, while the other variance still enters the denominator.

Lost-success AUROC and intervals. Among pairs whose clean run succeeds, the AUROC asks whether a larger response separates lost success (SF) from joint success (SS). Weighted SF–SS comparisons are averaged within each task, with half credit for ties, and eligible tasks are averaged equally. One shared set of 2,000 whole-task draws over the 28 tasks gives pointwise 95% intervals; an interval is suppressed with fewer than two eligible tasks, an undefined point, or fewer than 1,900 defined draws. Signals and layer were fixed; the estimates are post hoc.

What the table shows. All four signals rank with the command change under Sensor Noise, and P, W and AE under Camera; under Background and Light every interval includes zero. Only two correlations with dY exclude zero, Background AE and Camera W, both negative. Camera W, with AUROC $0.842[0.683, 0.967]$ on 10 tasks, is the largest of the 16 AUROC estimates (Sec. S4.1); the other Camera and Sensor Noise intervals include one half. Camera W thus has a positive command association and a negative signed-outcome association on the same pairs. It was not singled out in advance and rests on few tasks, so the main paper does not use it.

Attention output versus attention probabilities. The one contrast listed in advance, $\rho(W, \mathrm{command}) - \rho(P, \mathrm{command})$, shows neither association to be stronger: all four intervals include or touch zero (B $0.295[0.000, 0.592]$, C $0.119[-0.177, 0.430]$, L $-0.198[-0.648, 0.214]$, S $0.101[-0.124, 0.336]$). The ten-condition study instead uses partial ranks, with its own endpoint and support (Sec. S9.1).

Table S7. **Under Camera and Sensor Noise, L18 attention and action-expert responses rank with the command change; links to outcomes are weak.** Matched $\pi_{0.5}$ LIBERO-Plus pairs: 96/102/96/110 for Background, Camera, Light and Sensor Noise (B/C/L/S). Rows: the four fixed L18 signals at the first generation step. ρ: task-centered weighted rank correlation of the signal with the RMS of the first five commands (translation) or with the signed outcome dY (SF, lost success, $= -1$). AUROC: lost (SF) versus joint success (SS) among pairs whose clean run succeeds, over the eligible tasks in the last column; it conditions on clean success and the observed tasks. Brackets: pointwise 95% whole-task intervals. Background and Light have only three eligible tasks; under Background, prefix H has the same score in all eligible tasks, hence the degenerate interval $[0.833, 0.833]$.

Family	L18 signal	ρ with first five commands	ρ with signed dY	AUROC: SF versus SS	AUROC tasks
B	Prefix H	$-0.097[-0.397, 0.186]$	$-0.125[-0.243, 0.000]$	$0.833[0.833, 0.833]$	3
B	P TV	$-0.083[-0.357, 0.177]$	$-0.081[-0.352, 0.219]$	$0.333[0.000, 1.000]$	3
B	W	$0.212[-0.078, 0.492]$	$-0.178[-0.408, 0.101]$	$0.556[0.333, 1.000]$	3
B	AE	$0.015[-0.282, 0.296]$	$-0.275[-0.448, -0.067]$	$0.556[0.333, 1.000]$	3
C	Prefix H	$-0.072[-0.377, 0.245]$	$-0.185[-0.443, 0.107]$	$0.667[0.417, 0.896]$	10
C	P TV	$0.348[0.075, 0.596]$	$-0.076[-0.298, 0.164]$	$0.567[0.333, 0.788]$	10
C	W	$0.467[0.225, 0.679]$	$-0.284[-0.484, -0.044]$	$0.842[0.683, 0.967]$	10
C	AE	$0.400[0.116, 0.657]$	$-0.257[-0.475, 0.000]$	$0.700[0.455, 0.905]$	10
L	Prefix H	$-0.012[-0.262, 0.229]$	$-0.095[-0.233, 0.000]$	$0.611[0.500, 0.833]$	3
L	P TV	$0.175[-0.117, 0.468]$	$0.143[-0.047, 0.332]$	$0.222[0.000, 0.667]$	3
L	W	$-0.024[-0.344, 0.294]$	$-0.102[-0.280, 0.088]$	$0.722[0.500, 1.000]$	3
L	AE	$0.000[-0.302, 0.299]$	$-0.102[-0.291, 0.101]$	$0.528[0.250, 1.000]$	3
S	Prefix H	$0.620[0.359, 0.831]$	$-0.016[-0.297, 0.262]$	$0.463[0.194, 0.733]$	9
S	P TV	$0.453[0.174, 0.703]$	$-0.130[-0.360, 0.099]$	$0.593[0.333, 0.867]$	9
S	W	$0.554[0.312, 0.743]$	$-0.166[-0.393, 0.074]$	$0.630[0.394, 0.867]$	9
S	AE	$0.568[0.343, 0.757]$	$-0.036[-0.273, 0.193]$	$0.444[0.185, 0.692]$	9

S4.1. Influence of individual Camera tasks

Is the Camera attention-output AUROC carried by a single task? No: it stays above one half when any one task is removed (Table S8). Before deletion it is 0.842 (L18, first generation step), on ten eligible tasks with 15 lost and 23 joint successes.

Method. Each eligible task is removed once, and the table gives the average over the remaining tasks; SF–SS comparisons keep their setting and noise-draw weights, and tasks are weighted equally. The deletion rule was fixed before these calculations but after the association had been seen, so this is a post hoc influence analysis.

Table S8. **The Camera attention-output AUROC stays above one half after each single-task deletion.** All ten deletions; task indices follow the released table. Last column: exact mean after removing that task.

Task	SF	SS	Task AUROC	After deletion
1	1	1	1.000	0.824
2	1	3	1.000	0.824
3	2	2	0.750	0.852
4	1	3	0.667	0.861
5	1	3	0.667	0.861
6	1	3	1.000	0.824
7	2	2	1.000	0.824
8	1	3	1.000	0.824
9	2	2	1.000	0.824
10	3	1	0.333	0.898

S5. Command and Motion Changes Relative to a Clean Rerun

How large is a perturbation's effect on commands and motion, measured against a clean rerun? Under Sensor Noise, π_0 changes its first five commands and its end-effector position after five controls less than a clean rerun does, yet loses 14.29 points of success; under Camera, both policies end up further from their clean runs after 60 controls than a rerun does (Table S9). This supports Sec. 5.2 of the main paper: the size of a change is not its cost. Section S5.1 shows that command changes reach the robot.

The motion ratio is a ratio of task-equal means. Let R_0, R_1 be the clean runs with policy-noise draws 0 and 1, T_0, T_1 the perturbed runs with the same draws, and R'_0 a repeat of R_0 (main paper, Fig. 2d). TR is the mean of the two perturbed–clean distances and RR the clean-rerun distance. Both are averaged within tasks, then over tasks equally; we report $TR - RR$ and the ratio of task means $\overline{TR}/\overline{RR}$, below one when the perturbation changes less than a clean rerun. Unlike a mean of per-setting ratios, this ratio is not dominated by settings with a near-zero rerun distance; a zero RR leaves it undefined. A setting enters only if both perturbed–clean distances and the clean-rerun distance are measured in the same record (per policy, family, metric and control step), so numerator and denominator use the same settings. The repeat (R_0 vs. R'_0) and the two perturbed runs (T_0 vs. T_1) form separate panels with their own supports.

Missing outcomes are bounded, not imputed. The paired executions (Sec. S6) plan 268 settings for each of $\pi_{0.5}$ and π_0. All 536 policy–setting records exist, but 25 runs failed technically and 14 further outcomes failed validation. These 39 outcomes stay unknown in the fixed denominator of 2,680 runs. Rather than impute them, we report *bounds over missing outcomes*: the lowest and highest value an estimate can take over every assignment of success or failure to the unknown runs. A bound describes missing data, not sampling noise, which is reported separately. The primary scope is the 202 difficulty-labelled settings on the 28 common tasks per policy; all 268 settings form a sensitivity scope. For $\pi_{0.5}$ Light, the primary success change lies between 0 and $+8.04$ percentage points (-2.92 to $+5.21$ over all 268 settings); for π_0 Light it lies between -2.68 and $+5.36$ on the primary scope.

Measurements. The 15 metrics listed in advance are: the RMS differences of the first five commands (translation, rotation, gripper); end-effector position distance and orientation angle at controls 0/5/15/30/60; and two quantities at each run's own last step. The main paper's motion ratio uses position after five controls. Commands are in command units, positions in meters and orientations in radians. A fixed-time comparison uses only the pose stored at that control step.

π_0 under Sensor Noise moves less than a clean rerun and still loses success. For π_0 under Sensor Noise, the ratio to the clean rerun is $0.545[0.395, 0.718]$ for the first five commands and $0.580[0.409, 0.791]$ for end-effector position after control 5, while its primary success change is -14.29 percentage points. A family mean below the rerun level thus accompanies a net loss of success on the same settings. The position ratios after control 5 (main paper, Table 3) are 0.93 (Background), 2.68 (Camera), 0.57 (Light) and 1.32 (Sensor Noise) for $\pi_{0.5}$, and 0.62, 1.03, 0.24 and 0.58 for π_0. After control 60, Camera ratios exceed one in both policies, which also lose net success under Camera; this sets a distance at a fixed step against an outcome at the episode's end. For $\pi_{0.5}$ Background the ratio after control 60 is $1.273[0.9997, 1.6505]$, and the interval of its paired difference spans zero. Light's motion support is 43/48 settings on 26/28 tasks for $\pi_{0.5}$ and 44/48 on 27/28 for π_0.

Individual pairs against their own clean rerun. We also compare each perturbed pair of π_0 under Sensor Noise (primary scope) with the clean-rerun distance of its own setting. After control 5, let d_{vn} be the end-effector position distance between T_n and R_n for setting v and noise draw n, and let r_v be the distance between that setting's R_0 and R_1. Each recorded pair is classed by $d_{vn} < r_v$, $d_{vn} = r_v$ or $d_{vn} > r_v$, and all four outcome groups are kept (Table S10).

Most pairs that lose a success moved less than a clean rerun: of the 18 lost-success pairs, 12 have $d_{vn} < r_v$. In every setting with a lost-success pair, both clean runs succeed, so the clean rerun changes no outcome there. The twelve pairs cover ten settings and eight tasks; all lost-success pairs cover fifteen settings and ten tasks. All 55 settings on 28 tasks have both perturbed pairs, measured distances after control 5, a positive clean-rerun distance and known outcomes. The two pairs of a setting share one rerun distance, so the counts are nested within settings: they are a descriptive split, while the task-weighted estimates above keep the hierarchy.

Motion size is a poor predictor of lost success (post hoc AUROC). The main paper summarizes these pairs by an AUROC, computed post hoc: the probability that a lost-success pair has a larger ratio d_{vn}/r_v than a pair that keeps its success (ties count one half; one half is chance, one is perfect). The 18 lost-success pairs are the positive class and the 84 joint-success pairs the negative class, pooled without weights. The AUROC is 0.49, close to chance, with 95% interval $[0.29, 0.70]$: the percentile interval of 2,000 whole-task bootstrap draws that resample the 28 tasks with replacement and keep each task's pairs together, so the two pairs of a setting keep their shared r_v. Every draw contained both classes. With families pooled, the raw distance gives 0.73 ($\pi_{0.5}$) and 0.64 (π_0), and the norm of the change in the first five commands at the first query gives 0.74 and 0.64. Pooled values can reflect differences between families; comparing pairs only within their family gives 0.62 $[0.48, 0.74]$ and 0.55 $[0.35, 0.70]$ for the ratio. Per-family values: Sec. S6.

Outcome	$d < r$	$d = r$	$d > r$	Total
SS: both succeed	65	0	19	84
SF: lost success	12	0	6	18
FS: gained success	2	0	0	2
FF: both fail	4	0	2	6
All pairs	83	0	27	110

Table S10. **Pairs that lose a success and pairs that keep it both mostly move less than a clean rerun.** Counts for all 110 π_0 Sensor Noise pairs (55 settings on 28 tasks). Rows: outcome of the clean and the perturbed run (S, success; F, failure). Columns: whether the pair's end-effector position distance d_{vn} after control 5 is below, equal to or above its own setting's clean-rerun distance r_v, which the two pairs of a setting share. All distances and outcomes are available, and all 55 rerun distances are positive.

Details. The comparisons above are part of 720 listed in advance (two policies $\times$ two scopes $\times$ four families $\times$ fifteen metrics $\times$ three references). All are kept: 36 have no joint support and 684 have measured joint support. Each reports its exclusions, tasks, number of paired whole-task draws (2,000) and the reasons for undefined values. The 39 unknown outcomes enter the bounds on success change and on changed outcomes over the planned denominator; motion means use the measured joint support, and the SF/SS groups are post hoc. A zero command distance between R_0 and its repeat R_0' leaves that ratio undefined. Final poses are compared only when two runs end at the same control step, which is rare, so they form a separate, sparse comparison from control 60.

Table S9. **Relative to a clean rerun, Camera moves the end effector further after 60 controls in both policies, while several families change the commands less than a rerun does.** B, C, L, S: Background, Camera, Light, Sensor Noise. Columns: $TR - RR$ and the ratio of task means TR/RR for the first five commands (translation, command units) and for end-effector (EEF) position after control 60 (meters). TR, RR: perturbed–clean and clean-rerun distance. Joint N/tasks: settings with all three distances measured in the same record. Brackets: pointwise 95% intervals from paired whole-task draws on the support shown. Each of the 720 comparisons listed in advance keeps its own measured support.

Policy	Family	Joint N/tasks	Commands $TR - RR$	Commands TR/RR	EEF at control 60 $TR - RR$ (m)	EEF at control 60 TR/RR
$\pi_{0.5}$	B	48/28	$-0.0038[-0.0165, 0.0083]$	$0.91[0.66, 1.24]$	$0.00524[-0.000006, 0.01041]$	$1.27[0.9997, 1.65]$
$\pi_{0.5}$	C	51/28	$0.0572[0.0359, 0.0806]$	$2.36[1.80, 3.04]$	$0.02281[0.01480, 0.03160]$	$2.29[1.74, 3.03]$
$\pi_{0.5}$	L	43/26	$-0.0208[-0.0314, -0.0109]$	$0.52[0.38, 0.71]$	$-0.00104[-0.00598, 0.00326]$	$0.94[0.72, 1.24]$
$\pi_{0.5}$	S	55/28	$0.0115[-0.0050, 0.0281]$	$1.27[0.90, 1.76]$	$0.01298[0.00519, 0.02091]$	$1.77[1.26, 2.48]$
π_0	B	48/28	$-0.0343[-0.0535, -0.0159]$	$0.58[0.43, 0.77]$	$-0.00594[-0.01769, 0.00569]$	$0.83[0.57, 1.20]$
π_0	C	51/28	$-0.0017[-0.0212, 0.0181]$	$0.98[0.77, 1.27]$	$0.01924[0.00128, 0.04139]$	$1.55[1.04, 2.27]$
π_0	L	44/27	$-0.0619[-0.0799, -0.0452]$	$0.24[0.17, 0.35]$	$-0.01855[-0.02842, -0.00915]$	$0.48[0.36, 0.67]$
π_0	S	55/28	$-0.0373[-0.0551, -0.0204]$	$0.55[0.39, 0.72]$	$-0.00120[-0.01258, 0.00978]$	$0.97[0.70, 1.34]$

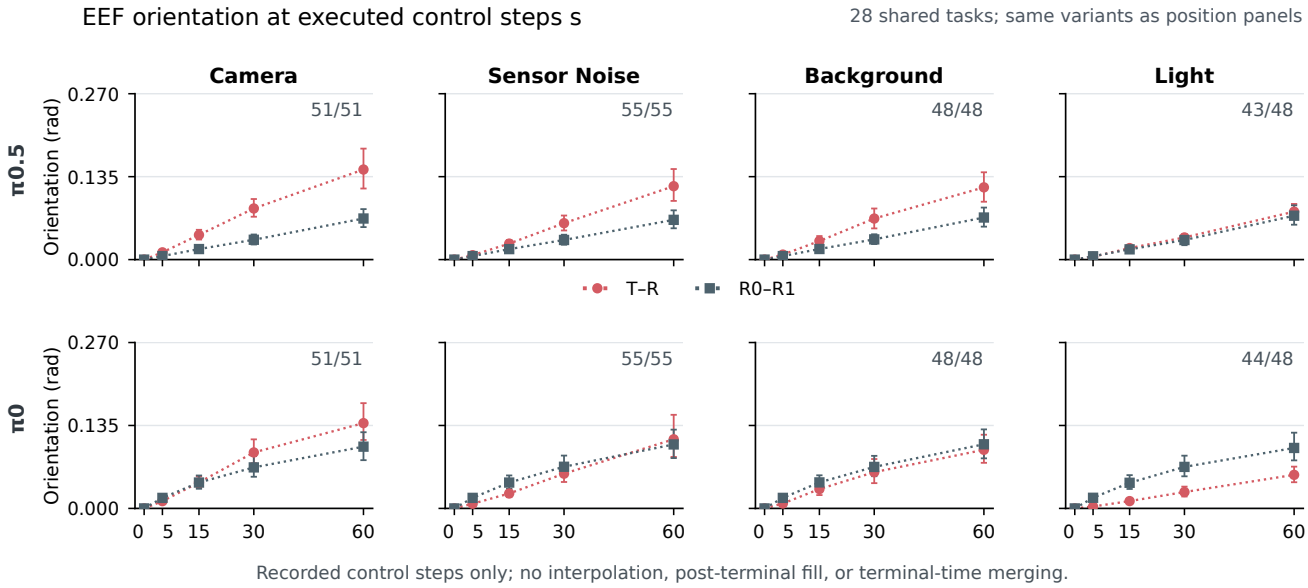


Figure S4. **End-effector orientation, a separate motion endpoint.** Perturbed–clean orientation distances in radians, with the same rule as the position panels (same setting, same control step); each control step has its own joint support. Lines connect the stored measurements; the two policies are shown in separate panels.

S5.1. Command changes reach the robot: strongly after 5 controls, still after 60

Do changes in the first five commands show in the robot's motion? The command change ranks with end-effector separation almost perfectly after five controls, as expected since those controls execute the commands, and more weakly but clearly after 60 (Table S11; Fig. S8). This supports Sec. 5.1 of the main paper. The comparison uses all 268 planned settings; its rules were fixed before the calculations, on data already observed.

Same pairs, two control steps. Both steps use the same pairs, so the drop from control 5 to control 60 is not a change of sample. A pair enters if its end-effector separation is valid at both steps; every outcome group is included. The matched support is 516/536 pairs for $\pi_{0.5}$ and 526/536 for π_0, each on 40 tasks; unavailable measurements stay counted on the planned support.

Task-weighted rank correlation. Pair weights $w_{tvn} = 1/(TV_tN_{tv})$ weight tasks, their settings and their noise draws equally at each level. For a scalar z, define the weighted midpoint empirical CDF

$$F_w(z_i) = \sum_{j:z_j<z_i} w_j + \frac{1}{2} \sum_{j:z_j=z_i} w_j. \qquad \text{(S4)}$$

The rank correlation is the weighted Pearson correlation of these transforms of command-change norm and end-effector distance, pooled over the plotted families. Unlike Sec. S4, ranks are not centered within tasks. Constant tasks stay included; zero overall variance would leave the statistic undefined.

Statistics and units. Intervals come from 10,000 whole-task bootstrap draws, shared by both control steps, with the weighted CDFs recomputed in each draw; percentile intervals use linear interpolation, and all draws are defined. Command norms use the fifteen translation coordinates (command units); end-effector distances are in millimeters.

Table S11. **The command change ranks with end-effector separation more strongly after control 5 than after control 60, in both policies.** Weighted rank correlation on the same pairs at both steps; the last column is their difference. Brackets are pointwise 95% intervals for each policy and contrast.

Policy	Control 5	Control 60	Early minus late
$\pi_{0.5}$	0.953 [0.933,0.968]	0.474 [0.362,0.580]	0.479 [0.368,0.593]
π_0	0.974 [0.961,0.982]	0.409 [0.303,0.512]	0.566 [0.463,0.671]

S6. Paired Task Outcomes against a Repeat and a Clean Rerun

Which runs lose success under a perturbation, which gain it, and how does that compare with a clean rerun? Lost successes exceed the clean-rerun level under Camera and Sensor Noise in both policies; gained successes exceed it in no family. Net success hides such changes: under Background, $\pi_{0.5}$ loses and gains successes equally, so its net success is unchanged while outcomes still change (Table S13). Camera costs net success in both policies, and Sensor Noise in π_0. This supports Sec. 5.2 and Fig. 5 of the main paper. Figure S9 gives the full comparison with the repeat and the clean rerun. These executions are separate from the internal recordings.

S6.1. Paired success, lost and gained successes

Design. Each policy–setting record holds five runs, R_0, T_0, R_1, T_1, R'_0, with policy-noise draws $0, 0, 1, 1, 0$, all from the setting's matched initial state: two clean runs, two perturbed runs that each share one clean run's noise, so that a perturbed–clean pair differs only in the image, and a repeat of R_0. R_0 versus R_1 is the clean rerun, the reference for a perturbation's effect. R_0 versus R'_0 is the repeat, which checks that runs are reproducible. Every qualified repeat reproduces R_0's end-effector position after 60 executed controls exactly, except in three settings (two in $\pi_{0.5}$, one in π_0), where the positions differ by 0.03 to 1.17 mm with identical outcomes; the source of these small differences is unknown. Each policy keeps its own definition of the policy noise, and every comparison is formed within one record. The episode budget is fixed within each comparison; in the spatial suite it is 220 controls for π_0 and 280 for $\pi_{0.5}$.

Coverage. The study plans 268 settings (536 noise pairs) per policy, 536 policy–setting records in all, and 2,680 run

Table S12. **The paired executions cover all planned settings; 39 of 2,680 run outcomes are unknown.** Counts on the planned support of 268 settings per policy, five runs each, no interval; unknown outcomes enter the bounds over missing outcomes.

Coverage	$\pi_{0.5}$	π_0
Recorded policy–setting records	268/268	268/268
All five outcomes validated	257	263
Validated run outcomes	1,314/1,340	1,327/1,340
Unknown outcomes	26	13
Technically missing	17	8
Failed validation	9	5

outcomes. Every planned record exists, including the six specified $\pi_{0.5}$ Light settings. Table S12 keeps the full denominator: 2,641 outcomes are validated and 39 unknown, 25 through technically missing runs and 14 through failed validation.

The primary comparison uses the 28 common tasks and the 202 settings per policy that LIBERO-Plus assigns a difficulty level: 48 Background, 51 Camera, 48 Light and 55 Sensor Noise. The families share tasks, and each keeps its own selected settings and difficulty distribution; results describe this panel.

With χ the probability of a changed outcome and ΔSR the success change, lost successes (clean success, perturbed failure) and gained successes (the reverse) give the paired identities

$$\begin{aligned} \chi &= \Pr(\text{lost}) + \Pr(\text{gained}), \\ \Delta\text{SR} &= \Pr(\text{gained}) - \Pr(\text{lost}). \end{aligned} \tag{S5}$$

Definitions. For binary outcomes Y_R, Y_T of a clean and a perturbed run, a lost success (SF) has probability $\Pr(Y_R = 1, Y_T = 0)$ and a gained success (FS) $\Pr(Y_R = 0, Y_T = 1)$, both unconditional paired probabilities under the stated weights. The success change (RD) is

$$\text{RD} = \text{SR}(T) - \text{SR}(R) = \Pr(\text{gained}) - \Pr(\text{lost}), \tag{S6}$$

and the rate of changed outcomes is their sum. The clean rerun (R_0 vs. R_1) has a rate of changed outcomes but no direction, since its two runs differ only in their noise draw; its *rerun level* for lost or for gained successes is half that rate. A rate exceeds the rerun level when the interval of its excess over that level lies above zero. Unknown outcomes take every possible value on the planned support, giving bounds over missing outcomes (Sec. S5), not sampling intervals.

Net success hides lost and gained successes, measured against the clean rerun. Under Background, $\pi_{0.5}$ loses and gains 3.57% of outcomes each: its success change is zero while 7.14% of outcomes change. The two Camera point estimates coincide; each policy has its own interval. The table below gives the rates of changed outcomes of perturbed–clean pairs and of the clean rerun, and the excess of the first over the second (task-equal percentages, complete records); a positive excess means more changes than policy noise alone. Under Camera and Sensor Noise, perturbations change more outcomes than a clean rerun in both policies; under Background they do not (Light is incomplete). Split by direction, lost successes exceed the rerun level under Camera and Sensor Noise in both policies (excess $+10.71$ to $+23.21$ points, lower interval limits $+0.89$ to $+11.61$), and gained successes exceed it in no family (-0.96 to $+2.68$, every interval including zero). Table S14 adds the repeat.

Policy / family	Perturbed	Clean rerun	Excess [95% interval]
$\pi_{0.5}$ / B	7.14	3.57	$+3.57$ $[-2.68, +8.93]$
$\pi_{0.5}$ / C	29.46	3.57	$+25.89$ $[+13.39, +39.29]$
$\pi_{0.5}$ / S	20.54	7.14	$+13.39$ $[+2.68, +25.00]$
π_0 / B	10.71	7.14	$+3.57$ $[-5.36, +12.50]$
π_0 / C	24.11	5.36	$+18.75$ $[+4.46, +32.14]$
π_0 / S	17.86	0.00	$+17.86$ $[+9.82, +26.79]$

Statistics. Estimates average the two perturbed–clean comparisons equally (each noise draw has weight one half), then settings within tasks, then tasks equally. The adjusted intervals of Table S13 and the excess intervals above use 2,000 whole-task draws with these fixed weights. Resampling whole tasks keeps both noise draws and all five runs together, so intervals describe variation over tasks given the recorded noise draws.

S6.2. Commands and motion on the same executions

Camera changes the commands and moves the end effector most in both policies; Light has the smallest means and incomplete support (Table S15). Supports follow Sec. S5; outcome estimates keep the full planned denominator.

End-effector positions at controls 0, 5, 15, 30 and 60 are read when the environment call returns; controller commands and contact histories are separate measurements. Final observations follow each run's own last step rather than a shared step; missing poses are neither interpolated nor replaced by final poses.

Internal states in the paired executions. The $\pi_{0.5}$ internal states stored here are recomputed (prefix H, the states of the context tokens, before each block; action-expert H after each block), unlike the states recorded during the policy's own generation in the internal recordings; their availability is checked per run. The archived π_0 runs stored no internal states; Sec. S14 regenerates first-query states for 369 pairs.

Table S13. **Camera costs net success in both policies, and Sensor Noise in π_0; under Background, $\pi_{0.5}$ loses and gains equally and its net success is unchanged.** Primary support of 28 tasks and 202 settings per policy. Success, lost and gained successes in percent; success change and its interval in percentage points. Bracketed estimates, including all incomplete Light estimates, are bounds over missing outcomes; the last column is the 98.75% whole-task bootstrap interval, Bonferroni-adjusted over the four families within each policy.

Policy/family	SR(R)	SR(T)	Lost	Gained	Success change	Adjusted interval for success change
$\pi_{0.5}$ / B	94.64	94.64	3.57	3.57	0.00	[-7.14,8.93]
$\pi_{0.5}$ / C	94.64	74.11	25.00	4.46	-20.54	[-39.29,-1.79]
$\pi_{0.5}$ / L	91.07	[91.07,99.11]	[0.00,5.36]	[5.36,8.04]	[0.00,8.04]	Not identified
$\pi_{0.5}$ / S	92.86	84.82	14.29	6.25	-8.04	[-23.21,5.36]
π_0 / B	89.29	85.71	7.14	3.57	-3.57	[-13.39,5.36]
π_0 / C	86.61	66.07	22.32	1.79	-20.54	[-35.71,-6.25]
π_0 / L	87.50	[84.82,92.86]	[1.79,6.25]	[3.57,7.14]	[-2.68,5.36]	Not identified
π_0 / S	92.86	78.57	16.07	1.79	-14.29	[-28.57,-2.68]

Table S14. **Perturbed runs change more outcomes than the repeat of the first clean run, which changes no validated outcome, under Background, Camera and Sensor Noise.** Population and weights as above; rates of changed outcomes (lost plus gained) in percent, brackets are bounds over missing outcomes, no intervals. Columns: perturbed run, repeat of the first clean run (same noise), clean rerun (different noise). Light is reported with its bounds.

Policy/family	Perturbed T vs. R	Repeat R_0 vs. R_0'	Clean rerun R_0 vs. R_1
$\pi_{0.5}$ / B	7.14	0.00	3.57
$\pi_{0.5}$ / C	29.46	0.00	3.57
$\pi_{0.5}$ / L	[5.36,13.39]	[0.00,10.71]	10.71
$\pi_{0.5}$ / S	20.54	0.00	7.14
π_0 / B	10.71	0.00	7.14
π_0 / C	24.11	0.00	5.36
π_0 / L	[5.36,13.39]	[0.00,8.93]	3.57
π_0 / S	17.86	0.00	0.00

Table S15. **Camera changes commands and moves the end effector most in both policies; Light has the smallest means and incomplete support.** Joint N/tasks: primary-panel settings with perturbed–clean and clean-rerun distances for that metric and step. Task-equal perturbed–clean means, no intervals. Translation RMS over the first five commands in command units; EEF5/EEF60 are end-effector position distances after controls 5 and 60, stored in meters, shown in centimeters.

Policy/ family	Joint N/tasks	Translation RMS	EEF5 (cm)	EEF60 (cm)
$\pi_{0.5}$ / B	48/28	0.03813	0.291	2.441
$\pi_{0.5}$ / C	51/28	0.09916	0.836	4.046
$\pi_{0.5}$ / L	43/26	0.02269	0.181	1.588
$\pi_{0.5}$ / S	55/28	0.05344	0.411	2.975
π_0 / B	48/28	0.04750	0.374	2.849
π_0 / C	51/28	0.08016	0.625	5.424
π_0 / L	44/27	0.01933	0.144	1.713
π_0 / S	55/28	0.04465	0.351	3.349

Otherwise the policies are compared on commands, motion and paired outcomes.

A family can change commands more yet cost less success. A ranking comparison listed in advance asks how often the order of two families reverses: one has the larger command or motion change, the other the larger net loss of success. All six ordered comparisons among Background, Camera and Sensor Noise are stored for the primary command measure; each fraction is the share of 2,000 resampled task sets in which the reversal holds strictly (ties count as none). For π_0, Background against Sensor Noise, the fractions are 69.9% for commands, 67.75% for EEF5 and 2.8% for EEF60, so the reversal depends on the measure and the control step.

Estimators that could not be computed. A planned estimator of the difference between correlations is undefined on these executions: for both policies and all five fixed primary command and end-effector metrics, none of the 2,000 draws defines it on the fixed support. The aggregation and lost-success comparisons that can be estimated use the studies in Secs. S9 and S10.4.

Motion size and lost success, by family (post hoc). The table below gives the AUROC of Sec. S5 (motion ratio after control 5; lost against joint successes) per policy and family, with lost/joint-success pair counts in parentheses; the last row uses motion after control 60. Light has too few lost-success pairs. New draws make the π_0 Sensor Noise interval differ slightly from Sec. S5.

Family	$\pi_{0.5}$	π_0
Background	0.39 [0.13, 0.68] (4/89)	0.67 [0.25, 0.90] (8/80)
Camera	0.57 [0.40, 0.73] (27/71)	0.57 [0.38, 0.74] (24/65)
Sensor Noise	0.74 [0.57, 0.89] (16/86)	0.49 [0.29, 0.69] (18/84)
Pooled	0.72 [0.62, 0.82] (47/328)	0.64 [0.48, 0.77] (51/309)
Pooled, control 60	0.60 [0.47, 0.72]	0.68 [0.57, 0.77]

S7. Repeatability of Command Changes across Policy-Noise Draws

Does a perturbation change the commands in the same way under a different draw of policy noise? For the arm it does: across the four families, the agreement of the translation changes is 0.884–0.920 in $\pi_{0.5}$ and 0.799–0.889 in π_0, where 1 means identical changes (Table S16). This supports Sec. 5.1 of the main paper: the first five commands are a repeatable readout of a perturbation. Gripper changes agree much less.

Rotation repeats nearly as well; the gripper does not. Rotation agreement spans 0.818–0.858 in $\pi_{0.5}$ and 0.719–0.822 in π_0. Agreement of the continuous gripper command is lower, 0.143–0.371 and 0.077–0.342. Ranges span the family point estimates; Table S16 gives the intervals.

The same perturbation under two noise draws. We use the paired executions (Sec. S6), where R_0/T_0 and R_1/T_1 share a policy-noise draw, so the two command changes of a setting differ only in that draw. All four runs need five commands at the first query ($q = 0$), with verified initial state, source, noise draw and command positions; a command stays missing if its environment call did not return. All outcome groups, including joint failure, are eligible.

A signed agreement score. The score divides the inner product of the two changes by their mean squared size: equal changes score one, unequal changes in the same direction less, and opposing changes below zero. Flatten the first five commands and write $d_n = T_n - R_n$, $n \in \{0, 1\}$; translation and rotation have 15 coordinates each and the gripper five. Define

$$D = \tfrac{1}{2}(\|d_0\|^2 + \|d_1\|^2), \tag{S7}$$

$$M = \left\|\tfrac{1}{2}(d_0 + d_1)\right\|^2, \quad V = \tfrac{1}{4}\|d_0 - d_1\|^2, \tag{S8}$$

$$U = d_0^\top d_1 = M - V, \qquad D = M + V. \tag{S9}$$

In compact notation, with $\delta u_n = d_n$ the change in the first five commands under noise draw n, the numerator and denominator are

$$U = \delta u_0^\top \delta u_1, \quad D = \tfrac{1}{2}\big(\|\delta u_0\|^2 + \|\delta u_1\|^2\big). \tag{S10}$$

The agreement for one setting is

$$s = \frac{U}{D} = \frac{2 d_0^\top d_1}{\|d_0\|^2 + \|d_1\|^2} = \frac{2r\cos\theta}{1 + r^2}, \quad s \in [-1, 1], \tag{S11}$$

where the last form requires both norms to be positive and $r = \|d_1\|/\|d_0\|$. If both changes are zero, s is undefined; if only one is zero, $s = 0$ while the cosine is undefined. No epsilon or clipping is used. Reading U as a squared population mean would require independent, identically distributed noise; the ratio is a descriptive score for the two recorded draws, not an unbiased estimator of a population ratio.

Support and weighting. The primary plan has 202 settings on 28 common tasks per policy. All four runs have the first five commands in 197/202 settings for $\pi_{0.5}$ and 198/202 for π_0; Light has 43/48 settings on 26 tasks and 44/48 on 27, while the other families keep all primary settings and 28 tasks (Table S16). The full sensitivity scope keeps 258/268 and 263/268 settings (stored full-scope table); its families originally cover 34 Background, 32 Camera, 40 Light and 39 Sensor Noise tasks. Command eligibility and outcome availability are counted separately; the 39 unknown outcomes stay in the denominator of 2,680 runs.

Statistics. Per-setting scores are averaged within task, then over tasks equally, per policy, family and component (a mean of ratios). The 2,000 whole-task bootstrap draws share their task resampling across series, keep settings and noise pairs nested, and are all valid; the 95% percentile intervals are pointwise and condition on the two noise draws and the selected settings.

Opposing changes and zero changes stay in the comparison. The continuous gripper changes in opposite directions ($U < 0$) in 36/197 and 62/198 primary settings, and the π_0 Light gripper interval crosses zero in both scopes. Translation never has $U < 0$; π_0 Sensor Noise rotation has it once. All scores are defined; no setting has only one zero change. In the full scope the gripper has $U < 0$ in 52/258 and 75/263 settings; this larger scope is a sensitivity analysis, not an independent replication.

Light has different eligible settings in the two policies, so a direct comparison of the policies needs their common settings. Section S7.1 reads the gripper command as an event instead.

S7.1. Gripper timing does not clearly improve prediction of lost success

Does the timing of the first opening-to-closing gripper command add information about lost success beyond the size of the command changes? At a cutoff of 60 controls, fixed in advance, adding timing lowers the held-out Brier loss slightly in both policies, with intervals that include zero (Table S17). This is an additional, exploratory study; the main paper does not rely on it. The feature is a command event, not physical closure or object contact.

Event reconstruction. Across all 50 runs, the Panda gripper command is positive for closing and negative for opening;

Table S16. **Translation and rotation changes repeat across policy-noise draws; the gripper agrees less.** Primary 28-task plan per policy; Settings and Tasks give defined/planned support. Entries are task-equal means of the per-setting agreement U/D with 95% whole-task intervals. These are normalized inner products of the changes under the two draws, not energy percentages.

Policy	Family	Settings	Tasks	Translation	Rotation	Continuous gripper
$\pi_{0.5}$	Background	48/48	28/28	$0.898\,[0.843, 0.946]$	$0.858\,[0.793, 0.913]$	$0.371\,[0.270, 0.467]$
$\pi_{0.5}$	Camera	51/51	28/28	$0.920\,[0.872, 0.957]$	$0.818\,[0.733, 0.892]$	$0.362\,[0.282, 0.442]$
$\pi_{0.5}$	Light	43/48	26/28	$0.884\,[0.810, 0.943]$	$0.848\,[0.778, 0.907]$	$0.143\,[0.009, 0.279]$
$\pi_{0.5}$	Sensor Noise	55/55	28/28	$0.909\,[0.870, 0.942]$	$0.849\,[0.788, 0.908]$	$0.371\,[0.250, 0.491]$
π_0	Background	48/48	28/28	$0.865\,[0.810, 0.909]$	$0.807\,[0.720, 0.877]$	$0.342\,[0.219, 0.453]$
π_0	Camera	51/51	28/28	$0.889\,[0.840, 0.928]$	$0.814\,[0.756, 0.866]$	$0.325\,[0.185, 0.460]$
π_0	Light	44/48	27/28	$0.799\,[0.711, 0.880]$	$0.719\,[0.633, 0.797]$	$0.077\,[-0.090, 0.245]$
π_0	Sensor Noise	55/55	28/28	$0.844\,[0.780, 0.896]$	$0.822\,[0.762, 0.876]$	$0.228\,[0.078, 0.388]$

Table S17. **Adding gripper timing to the command-change sizes does not clearly lower the loss at a fixed cutoff of 60 controls.** Cutoff pairs (of 404 per policy) reach the cutoff with valid command records and outcomes; the subset whose clean run succeeds (N/tasks) is analyzed. Improvement is baseline minus augmented out-of-fold Brier loss, positive when timing lowers the loss, with a 97.5% paired whole-task interval. The denominator 404 is the fixed plan per policy; 375/360 pairs are analyzed.

Policy	Cutoff pairs	Clean-success N/tasks	Baseline Brier	Augmented Brier	Improvement [97.5% interval]
$\pi_{0.5}$	396/404	375/27	0.09664	0.09555	$0.00109[-0.00224, 0.00448]$
π_0	397/404	360/26	0.12323	0.12172	$0.00151[-0.00471, 0.00724]$

zero leaves the accumulated command unchanged. Events use only complete, finite command records with valid identifiers up to the cutoff; the 39 unknown run outcomes, early terminations and technical failures are kept.

Baseline and augmented predictors. Of the 404 primary pairs per policy, both runs must reach the cutoff with valid command records and outcomes, and the clean run must succeed. Both predictors use only information available at the cutoff. The baseline uses $\log(1 + \mathrm{RMS})$ of the translation, rotation and continuous-gripper command differences over the same first 60 controls, plus family indicators. The augmented predictor adds whether each run has an opening-to-closing transition by the cutoff, the difference of their times divided by 60 when both do, and each run's initial command direction; occurrence flags separate an absent event from a zero time difference.

Fitting on held-out tasks and intervals. The ridge model, fixed in advance, uses penalty $\alpha = 1$, an unpenalized intercept, centering and scaling on the training data only, and leave-one-task-out prediction clipped to $[0, 1]$; tasks are weighted equally, and families, settings and noise draws equally within their parents. Both predictors use identical rows in each fold; no training fold has a single class (which would fall back to the training mean). Both initial-closing indicator columns are zero in all eligible rows and add no variation. Paired whole-task resampling (2,000 draws) gives 97.5% intervals per policy; the predictors are not refitted inside the bootstrap.

Outcomes of the analyzed pairs. On the primary support the outcome counts are $328/47/17/4$ SS/SF/FS/FF with eight unknown pairs for $\pi_{0.5}$ and $309/51/10/27$ with seven unknown pairs for π_0; every validated pair is kept whether or not a transition occurs. Pairs whose clean run succeeds number $93/98/82/102$ (Background/Camera/Light/Sensor Noise) for $\pi_{0.5}$ and $88/89/81/102$ for π_0. The estimates in Table S17 apply to this conditional support.

S8. Response Structure at a Later Observation

Does the response keep its shared structure later in an episode? Shared directions fitted at one mid-episode observation retain far more of the response at a later observation than clean-feature or random directions (Sec. S8.1). This supports main Sec. 4.1, though here only tasks are held out.

S8.1. Shared directions fitted at one observation hold at a later observation

Directions fitted at an earlier observation retain 50.8–64.2% (L1), 35.7–45.7% (L9) and 32.7–38.1% (L18) of the response energy around the fitting-data mean at a later observation on 16 held-out tasks, against at most 11.0% for clean-feature and 1.8% for random directions, in all 30 condition–layer comparisons (Table S18). Tasks are held out but conditions are not: the test tasks are never used in fitting, while the ten test conditions are.

Data: two observations per episode. Each episode is observed twice, so that directions fitted at one state are tested at another. The study runs $\pi_{0.5}$ on original LIBERO under ten persistent conditions, which stay on for the whole episode (Fig. S5 legend): 3,360 episodes with two observations each, 6,720 records. Features exist for 6,554 records; 83 missing episodes (166 records) stay missing. The two observations are taken after a suite-specific number of executed controls, Spatial 20/40, Object 25/55, Goal 30/60 and LIBERO-10 50/100, with no feature recording in between. At each observation, clean and perturbed images of the same visited state pass through the policy with the same policy noise; we record clean features and responses at L1, L9 and L18, first generation step ($j = 0, t = 1$), first five planned actions ($3 \times 5 \times 1024$ per observation). Between the observations, state, images and noise all change.

Fit: held-out tasks, seen conditions. Of the 40 tasks, 24 are used for fitting and 16 are held out. Only the first observations of the fitting tasks enter the fit, 1,404/1,440 available observations. Tasks, conditions, start states, noise repeats and the five plan positions are each weighted equally within the level above. Each layer has its own mean μ_p for each position p and a channel scatter shared across positions,

$$C = \sum_{i,p} \frac{w_i}{5} (\Delta H_{ip} - \mu_p)^\top (\Delta H_{ip} - \mu_p). \qquad \text{(S12)}$$

Its top 16 eigenvectors form the shared directions $B \in \mathbb{R}^{1024\times16}$; the largest absolute loading of each fixes its sign. The rank is fixed and nothing is selected by outcome. Clean-feature directions use the same observations and weights but are fitted to clean features; random directions are a fixed Gaussian basis orthonormalized by QR. All three reconstruct the same response about the same μ_p, with 16 coefficients per position computed from the test response, 80 per observation out of 5120 response entries, so they differ only in their directions. A mean-only baseline uses μ_p with zero coefficients.

Measures. For one observation, let $E = \|\Delta H\|_F^2$, $E_c = \|\Delta H - \mu\|_F^2$ and $S_a = \|\Delta H - \widehat{\Delta H}^{(a)}\|_F^2$ be the raw energy, the centered energy and the reconstruction error of directions a. The plotted centered retention and the tested advantage over a control (error saved, as a share of raw energy) are

$$R_a = 1 - \frac{S_a}{E_c}, \qquad G_{\text{response}-a} = \frac{S_a - S_{\text{response}}}{E}. \qquad \text{(S13)}$$

R_a pools all five positions' energy before forming the ratio. Negative values of G are kept, and ratios with a zero denominator are undefined. Channel participation, the effective number of channels carrying the response energy, describes its spread; it is not a spectral rank or an intrinsic dimension.

The advantage over both controls holds in every comparison. All 30 advantages have positive 95% lower bounds (descriptive). Retention changes little between the two observations: for the paired first-to-second-observation difference in retention of the shared directions, 28/30 intervals cross zero; the remaining two are pointwise contrasts without multiplicity adjustment.

Retention holds even when the response changes direction. Fig. S5 checks that retention does not merely reflect an unchanged response; it keeps all 960 planned held-out episodes (940 available, 20 missing). Its horizontal axis is the cosine between the full 5×1024 responses at the two observations. Even near zero cosine, the shared directions retain appreciably more energy than clean-feature and random directions: the response changes, but it stays within the shared span. The scatter is descriptive, with no fitted threshold or association model.

Statistics and implementation notes. Intervals come from 2,000 whole-task bootstrap draws, which keep the dependence within tasks and among comparisons on the same 16 tasks, and condition on the fitted directions and the observed support. The random directions and the bootstrap use a fixed seed (20261012).

Table S18. **At the later observation, shared directions retain far more of the response than both controls; retention falls with depth while participation rises.** Columns: centered retention (higher is better) of three sets of 16 directions, then channel participation. 16 held-out tasks, ten persistent conditions; each cell has 94/96 defined observations and 16/16 tasks. Ranges are over condition means, averaging available noises within start states, start states within tasks, and tasks equally. All three reconstruct the same position-centered response (plan-position means removed). Participation pools position energies on 1024 channels.

Layer	Shared directions	Clean-feature directions	Random directions	Channel participation
L1	50.8–64.2%	3.6–6.3%	1.6–1.8%	31.0–39.9
L9	35.7–45.7%	0.8–1.8%	1.4–1.8%	63.9–117.9
L18	32.7–38.1%	8.2–11.0%	1.4–1.5%	186.2–229.0

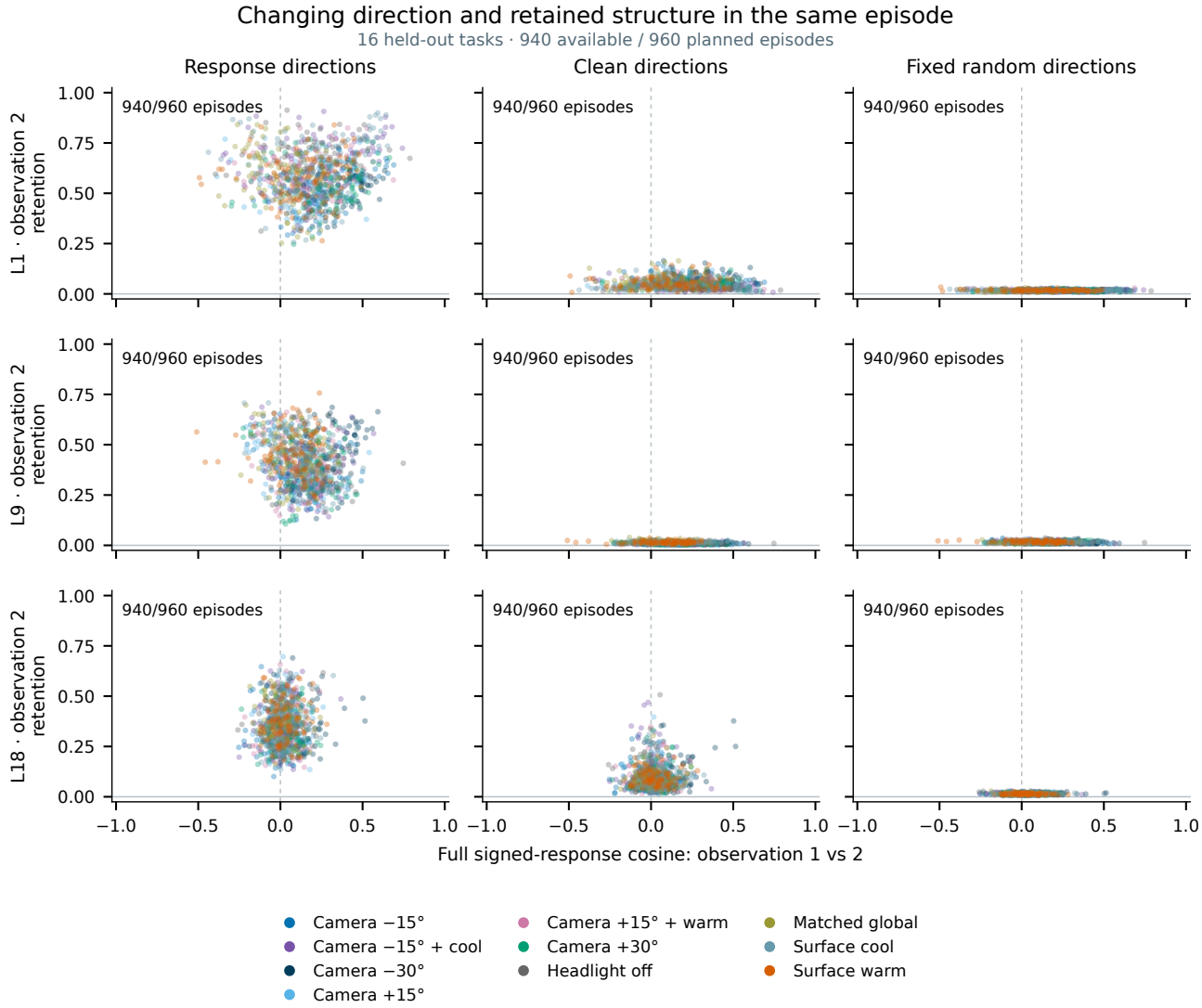


Figure S5. **Shared directions fitted at an earlier observation retain more of the later response than both controls, even when the response changes direction.** Rows: layers. Columns: shared, clean-feature and random directions (titled Response, Clean, Fixed random), with equal coefficient budgets. Each point is one episode: cosine between its full responses at the two observations (horizontal) against centered retention at the second (vertical); colors mark conditions. Sec. S8.1: 16 held-out tasks, 940/960 planned episodes; no interval.

S9. Further Studies on Original LIBERO

Two studies of $\pi_{0.5}$ on original LIBERO, each with its own tasks, start states and unit of analysis, show that a larger command change need not mean a larger loss of success (main Sec. 5.2). The command change orders ten conditions but not task–condition cells (Sec. S9.1); headlight removal and a matched dimming change the commands very differently yet succeed equally often (Sec. S9.2).

S9.1. Command change ranks conditions by success change, but not task–condition cells

A condition is one visual change applied to all 40 tasks; a cell is one task under one condition, with up to six pairs (two start states, three policy-noise draws). Across the ten condition means, mean command change ranks mean success change with rank correlation -0.903 $[-0.952, -0.309]$: conditions that change the commands more lose more success. Across the 400 cells, the correlation is -0.075 $[-0.195, +0.043]$, an interval that includes zero (Table S19). A condition mean averages over tasks and keeps only differences between conditions; cells also carry within-condition variation, where the command change says little about success (below). This supports main Sec. 5.2; unlike there, the command change is measured mid-episode.

Data: 2,344 of 2,400 planned pairs. The study runs ten persistent visual conditions (on for the whole episode) on 40 tasks. Paired internal measurements and outcomes are available for 2,344/2,400 planned perturbed/clean pairs: 2,110 joint successes, 104 lost successes, 83 gained successes and 47 joint failures. Of the 400 cells, 364 have six pairs, 20 have four and 16 have five; missing pairs remain missing. The 236 observed clean episodes serve all ten conditions with their original labels.

What is measured, and when. Commands and features are taken after two scheduled numbers of controls: 20/40 for Spatial, 25/55 for Object, 30/60 for Goal and 50/100 for LIBERO-10. At each, the perturbed run is compared with a clean evaluation of the same state at the first generation step; the two values are averaged within each episode, and each outcome is counted once. The command change uses the first five planned commands (translation part, 15 numbers). The L18 response magnitude is the relative RMS over five positions and 1,024 channels. Main-camera conditional total variation (TV), attention redistribution within main-camera tokens, requires all eight heads. Condition means weight episodes equally within condition; cell means are ranked equally.

Table S19. **The command change ranks condition means by success change, but not task–condition cells.** First two rows: rank correlation between the mean command change (translation part) and the mean success change, across the ten condition means and across the 400 task–condition cells; negative: larger command change, larger loss. Last three: lost-success AUROC (0.5 is chance), comparing lost with kept successes within conditions and weighting SF–SS pairs, unlike the task-equal AUROC of Sec. S4. Ten-condition study, 2,344 perturbed/clean pairs, 40 tasks; each scalar is averaged over two observations; estimates use case-level scalars; intervals in the text.

Comparison	Estimate
Command change–success change, condition means	-0.903
Command change–success change, task–condition cells	-0.075
L18-response lost-success AUROC	0.535
Main-camera conditional-TV lost-success AUROC	0.581
Command-change lost-success AUROC	0.421

Within conditions, response size barely separates lost from kept successes. The lost-success AUROC asks how well a measure ranks lost successes (SF) above kept ones (SS) within conditions, weighting the 21,405 SF–SS comparisons, not tasks. All three are near chance (0.5); the command-change AUROC is below it, so within a condition, pairs that lose a success have, if anything, smaller command changes.

After removing task and condition effects, the internal measures co-vary but hardly predict success. We rank each variable globally with average ties and remove additive task and condition effects, so that these cannot drive the correlations. Pearson correlation of the residual ranks is 0.722 between L18 response and command change, 0.401 between L18 response and main-camera TV, and 0.286 between main-camera TV and command change. Their correlations with the success change ΔY are -0.0009 $[-0.0422, 0.0395]$, -0.0280 $[-0.0694, 0.0194]$ and $+0.0542$ $[0.0093, 0.0998]$ for L18 response, main-camera TV and command change. They describe association in the observed data.

Attention mass and attention shape carry different associations. A separate first-query comparison on the same 2,344 pairs centers percentile ranks within condition. The signed change of main-camera attention mass correlates -0.00351 with the command change; a Jensen–Shannon (JS) summary of the attention distribution correlates $+0.493$. Both average the first five positions over ten generation steps, three layers and eight heads; unlike conditional TV, JS pools layers. Both are point estimates without intervals.

Statistics and implementation notes. Intervals use 1,000 draws of the 40 tasks with replacement (seed 20261008), each carrying its start states, noise draws, conditions and shared clean episodes (for residual ranks, also missing pairs). The two rank correlations share these draws, recompute ranks and use linearly interpolated percentiles; residual-rank intervals have no suite stratification or multiplicity adjustment. Commands are in adapter coordinates; acknowledgements of execution are recorded separately.

S9.2. Headlight removal and a dimming matched in image statistics

We compare headlight removal with a global dimming of the lights, set per case so that each camera's first image matches the headlight-off image in mean intensity and mean absolute difference from clean. Removing the headlight changes the first five commands 5.12 times as much, yet both succeed on 77/80 cases, each on two cases where the other fails (Fig. S6). This supports main Sec. 5.2: a fivefold command change leaves the success count unchanged, and equal counts hide changed outcomes.

Fivefold command change, equal success counts, exchanged failures. The command change is the RMS of the 15 translation coordinates of the first five commands, perturbed minus clean. Its mean is 0.0570 for headlight removal and 0.0111 for matched dimming (ratio of means 5.12); the paired difference is 0.0459 $[0.0363, 0.0562]$ (95% interval over task clusters), and the headlight change is larger on 76 cases and smaller on four. The paired success difference is 0 $[-5, 5]$ pp; this interval describes the observed comparison, and an equivalence test was not prespecified.

Data and runs. The 80 cases come from 40 tasks and start states 10/11; each is run clean, with the headlight off and with matched dimming. Each run succeeds on 77 of 80 cases, and each pair of runs shares one failure (75 joint successes, 2 and 2 single failures, 1 joint failure). The runs share the start state and ten settling steps, checked before the first query.

The dimming is matched on initial image statistics only. Initial RGB images have shape $256 \times 256 \times 3$. A gain $g \in \{0.25, 0.275, \ldots, 0.75\}$ is chosen to minimize the largest of the four statistic mismatches (two statistics, two cameras), ties going to the larger gain. Dimming scales the ambient, diffuse and specular RGB of the active lights, including the headlight, and uses no policy output or outcome. All 80 cases meet the criterion of five code values and stay in the analysis: the largest mismatch is 3.94, gains lie in [0.35,0.55], and each gain stays fixed for the episode. Spatial lighting patterns and later images can still differ.

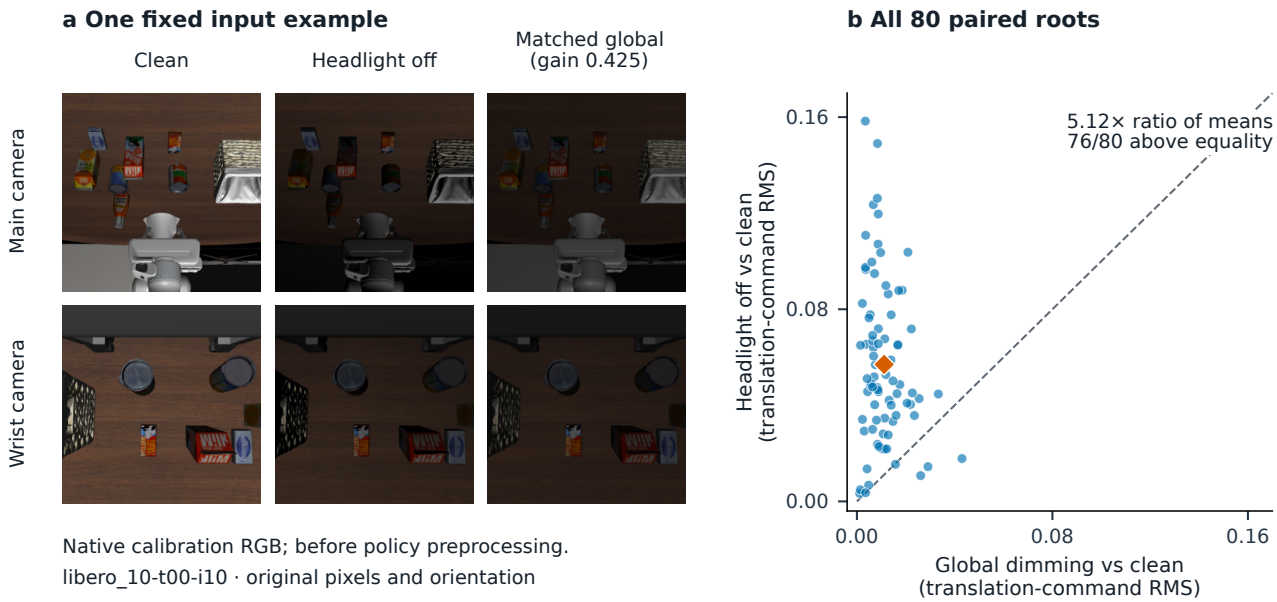


Figure S6. **Images matched in summary statistics can still change the commands differently.** (a) Main and wrist images of the first complete case in fixed order (LIBERO-10 task 0, start state 10, global gain 0.425), chosen by order, not by response or outcome; pixels and orientation as recorded. (b) All 80 cases: command change (RMS over 15 translation coordinates, in command units rather than displacement) under headlight removal against matched dimming; dashed: equality.

S10. Relating LIBERO-Plus Responses to Commands and Task Outcomes

This section describes the $\pi_{0.5}$ LIBERO-Plus recordings to which the shared directions of main Sec. 4.1 are fitted (first subsection); the rest is an additional study that the main paper does not rely on. Camera and Sensor Noise cost success; perturbed runs end farther from their clean runs than a clean rerun does, even when both succeed; and a prespecified test of whether response size identifies lost successes is inconclusive. This is consistent with main Sec. 5.2, whose outcome figures use the separate paired executions (Sec. S6); related analyses: Secs. S3.4 and S4.

S10.1. Task outcomes of the recorded pairs

The recordings: 536 pairs on 40 tasks. The recordings contain 268 settings, each under two policy-noise draws: 536 clean–perturbed pairs on 40 tasks in the LIBERO-Plus categories Background Textures, Camera Viewpoints, Light Conditions and Sensor Noise. Two inspected Light settings also change texture assets, so Light denotes the benchmark's prescribed settings, not an isolated lighting factor. Each pair has its own executed clean run; the 80 clean-run recipes fix only initialization and noise settings, not 80 trajectories shared across pairs. Outcomes (clean→perturbed) are 447 joint successes (SS), 56 lost successes (SF), 20 gained successes (FS) and 13 joint failures (FF). Pairs are nested within tasks.

The common-task comparison: 404 pairs on 28 tasks. All quantities below use one population, so that they are directly comparable: the 28 tasks shared by all four families, without 18 Light pairs lacking a LIBERO-Plus difficulty level, leaving 202 settings and 404 pairs, those of main Fig. 3 (Table S20). The estimates describe the selected settings; full-panel summaries and full-benchmark rankings refer to different populations.

Table S20. **Pairs per family in the full panel and on the common tasks.** $\pi_{0.5}$ LIBERO-Plus recordings: pairs/tasks in the full panel, pairs/settings on the common tasks, and common tasks per family. The full Light panel includes 18 pairs without a difficulty level. The common tasks are shared, but families still differ in settings, difficulty levels and number of settings per task.

Family	Full: pairs/tasks	Common: pairs/settings	Tasks
Background	118/34	96/48	28
Camera	112/32	102/51	28
Light	152/40	96/48	28
Sensor Noise	154/39	110/55	28

Camera and Sensor Noise cost success. On the common tasks, the success change (perturbed minus clean) is $+2.679$, -17.857, $+2.679$ and -12.500 percentage points for Background, Camera, Light and Sensor Noise, with pointwise 95% intervals $[-5.357, 11.607]$, $[-30.379, -6.250]$, $[-7.143, 13.393]$ and $[-22.321, -4.464]$, and simultaneous 95% intervals over the four families (Bonferroni-adjusted percentiles) of $[-7.143, 13.393]$, $[-34.821, -3.571]$, $[-8.929, 16.071]$ and $[-25.893, -2.679]$. The Camera and Sensor Noise intervals exclude zero; those of Background and Light, with positive point estimates, include both gains and losses. Each change is measured against the family's own clean runs and mix of difficulty levels. Estimates average the two noise draws within a setting, settings within a task–family, and then tasks equally; intervals use 10,000 bootstrap draws of whole tasks.

S10.2. Definitions of the response and attention measures

Absolute response size measures a difference, relative size compares it with the clean tensor, and attention total variation (TV) measures a change in how attention is allocated.

Recording sites. The first policy query is $q = 0$; j indexes generation steps and p plan positions. We record L1, L9 and L18 (zero-based block indices 0, 8, and 17). Prefix measurements replay the prefix computation with masks for valid tokens and defined queries. H is the block input and Z

the pre-attention RMSNorm output supplied to attention, so a larger H response at depth need not mean a larger change in what attention receives. Action-expert (AE) hidden states are recorded during the policy's own generation and keep the ten ordered plan positions at each generation step. Channel indices belong to their own representation and layer; equal indices in different spaces do not imply a shared direction.

Absolute and relative responses can move in opposite directions. For a selected set of N scalar coordinates, let X_R and X_T denote the clean and perturbed tensors and $\Delta X = X_T - X_R$. We report

$$d_{\mathrm{RMS}}(X) = \|\Delta X\|_F / \sqrt{N}, \tag{S14}$$

$$r(X) = \|\Delta X\|_F / \|X_R\|_F. \tag{S15}$$

Here the absolute AE L18 response falls from the first to the last generation step in every family–suite cell, while the relative response rises, also in every cell, because the clean norm falls. A relative trend therefore mixes response size with the clean scale. Ratios are formed per pair before averaging and are undefined for a zero clean norm; a ratio of family-average norms is a different estimator.

Attention change is measured over all visible keys. The attention probabilities P are not stored during the policy's run but recomputed at the recorded generation states. For each valid action query and attention head, the total variation over all visible keys is

$$\mathrm{TV}(P_T, P_R) = \tfrac{1}{2} \sum_{k \in \mathcal{V}} |P_T(k) - P_R(k)|, \tag{S16}$$

where $\mathcal{V}$ is the common visible-key set. We average over valid queries and heads and keep the probability mass on visual, text and other visible keys. The regional net mass change sums the signed probability change in a region; the regional conditional-distribution change first normalizes each run by its regional mass. Neither substitutes for full-key TV. The text region contains text tokens; the recorded configuration does not append the numerical joint state as discrete text.

Attention output combines allocation and transformed values. The content readout recomputes the attention output from prefix and action-side values, after projection, before the residual gate. Values and their weighted combination can change even when probability shifts decrease. This recomputation measures each branch at its saved generation state, separately from the recording of the policy's own generation (Table S1).

Commands and motion keep their own units. The output record separates the 32-coordinate policy actions from the seven-coordinate LIBERO commands after the adapter. We score translation, rotation and gripper components over the first five and over all ten plan positions. Command RMS uses saved command units; end-effector (EEF) distance uses the simulator's length units, without conversion to meters or millimeters. Orientation distance is the quaternion geodesic angle in radians. Gripper-state distance is the Euclidean difference of two recorded joint coordinates, not an opening width.

S10.3. Motion compared with the clean rerun

After 60 controls, perturbed runs are farther from their clean runs than a clean rerun is in every family, and under Camera and Sensor Noise also when both runs succeed. Two references give the separation a scale: the clean rerun, which changes only the policy noise, and a repeat of the clean run with the same inputs and noise.

Command differences are followed by motion. All 404 common-task pairs start at zero EEF-position, orientation and gripper-state distance, and all issue their first five commands. After 5 controls, the task-equal EEF distances are 0.002911, 0.008362, 0.002027 and 0.004108 for Background, Camera, Light and Sensor Noise, with pointwise intervals above zero. After 60 controls the means are 0.024468, 0.041819, 0.018961 and 0.030180. Each family keeps all 28 tasks over this window; later differences also reflect the different states the runs reach.

After 60 controls, separation exceeds the clean rerun. Let $\overline{d}_{TR,f}(60)$ be the task-equal clean–perturbed EEF distance for family f and $\overline{d}_{RR,f}(60)$ that between the two clean runs. After 60 controls their ratios (above 1: farther apart than a clean rerun) are 1.51, 2.39, 1.17 and 1.68 for Background, Camera, Light and Sensor Noise. Each ratio divides two family means, not an average of per-task or per-setting ratios, and has no estimated interval. The paired distance-difference intervals (2,000 bootstrap draws of the fixed 28 tasks, pointwise 95%) are in the released analysis package.

Extra motion and net success measure different effects. Background has a positive interval for motion beyond the clean rerun, yet its success interval includes zero. With the repeat as reference, the Background/Camera/Light/Sensor Noise reference distances are 0.002518/0.001873/0.002512/0.002565 on 28/26/26/28 tasks; the two references keep their own populations and means.

Runs that both succeed still move apart. Outcome groups refer to the case's own clean–perturbed pair: ALL

includes SS, SF, FS and FF, and SS and SF use that pair's final outcomes, not those of every repeat. After 60 controls, with the clean rerun as reference, Camera has $d_{TR} = 0.04360$ in SS (75 cases, 25 tasks) and 0.04291 in SF (23 cases, 12 tasks); Sensor Noise has 0.02872 in SS (91 cases, 27 tasks) and 0.03456 in SF (16 cases, 10 tasks). The SS intervals for motion beyond the clean rerun are positive for Camera, 0.02619 $[0.01678, 0.03596]$, and Sensor Noise, 0.01047 $[0.00181, 0.01859]$, while those of Background, 0.00730 $[-0.00019, 0.01414]$, and Light, 0.00228 $[-0.00348, 0.00756]$, include zero. These outcome-defined groups have different support. Both runs can thus succeed along different paths. Beyond 60 controls, finished episodes shrink the paired set, so later values need their task and case counts.

The two references. The repeat matches the recorded initial inputs, configuration, random-number state and policy noise; for each case it averages all eligible, independently executed repeats, whose remaining variation need not be pure floating-point error. The clean rerun compares the two policy-noise draws within each setting. All 202 settings support the clean-rerun comparison; the repeat supports 396 of the 404 cases. Both references keep the case's clean–perturbed pair.

S10.4. Does response size identify lost successes?

This prespecified test asks whether response size ranks lost successes (SF) above joint successes (SS) within task. Its adjusted intervals cross the thresholds of both prespecified criteria, for discrimination and for its absence, so it supports neither conclusion; the secondary pooled AUROCs are modest. This is consistent with main Sec. 5.2 but does not by itself establish it.

Four directed scores fixed before the test. The comparison starts from all 518 labelled pairs, not only the 28 shared tasks, and uses four fixed scores: the relative AE L18 response over the ten plan positions at the first generation step, the relative prefix main-camera L18 response, the L18 full-key attention TV at the first generation step, and the RMS change of the first five commands (translation). Larger values always rank as a more likely lost success; this direction was fixed in advance and kept after the results were seen.

Within-task discrimination with prespecified criteria. The primary AUROC compares SF with SS within each task, with half credit for ties, weighting eligible tasks equally; a task lacking either outcome has no primary AUROC. Camera contributes ten eligible tasks with 15 SF and 23 SS cases; Sensor Noise 13 with 18 SF and 32 SS cases. The positive criterion requires the same score's adjusted lower bound to exceed 0.60 under both Camera and Sensor Noise; the criterion of no practical discrimination requires all eight adjusted upper bounds to lie below 0.70. The 98.75% intervals for this restricted support (20,000 whole-task bootstrap draws shared across quantities and families, Bonferroni-adjusted over the four scores within each family) are in the released analysis package.

A secondary pooled comparison uses different support. The pooled secondary analysis keeps all SF and SS cases, takes the logarithm of each score with a fixed 10^{-12} floor and subtracts its task–family mean, so that task and family differences do not drive the result, before computing a pooled AUROC. Camera has 25 SF and 83 SS cases, with AUROCs 0.623/0.636/0.587/0.648 for the four scores; Sensor Noise has 22 SF and 125 SS, with 0.548/0.527/0.581/0.608. The adjusted intervals of the command score are [0.501,0.817] and [0.408,0.802]. Comparing higher with lower LIBERO-Plus difficulty levels under Camera and Sensor Noise, several responses increase while within-task success-change intervals cross zero.

S11. Removing the Shared L9 Directions without a Clean Observation

Restoration (Sec. S12) needs a clean observation of the same state. This test needs none: instead of copying clean values back at every generation step, it removes part of the L9 state's component along the shared directions, measured from a clean training mean, only at the first generation step of each query. In $\pi_{0.5}$ on original LIBERO (6,048 planned rollouts), the edit lowers the response at L18 by 8.8% under a camera change and 7.6% under headlight removal. Clean-feature directions with the same mean edit size raise it ($-2.5/-3.6\%$ reduction), and random directions lower it by 0.1–0.3%. It changes the first five commands by a small fraction of the perturbation's change and does not measurably change task success against the unedited policy. This supports the second test in main Sec. 4.2.

Design. The edit $H' = H - \lambda[(H - \mu_H)B]B^\top$ removes a fraction λ of the action-expert L9 state's deviation from the clean training mean μ_H along the 16 directions B, at all ten plan positions and only at the first generation step of each policy query; later steps and all other layers run unchanged. The directions are the L9 shared directions of the cross-observation study (Sec. S8.1), not the LIBERO-Plus fits. The controls use clean-feature or random directions with the same training-set mean edit norm over the three conditions, so that they differ in direction, not amount; this is Shared-large, and Shared-small uses half that dose. The design has 16 held-out tasks $\times$ 6 start states $\times$ 3 policy-noise draws $=$ 288 cases; each case runs 3 visual conditions (clean, Camera

$+15°$, headlight off) $\times$ 7 configurations (Unedited, Shared-small, Shared-large, Clean-feature, Random-1/2/3) = 6,048 planned main episodes, plus 16 technical repeats. Table S22 defines each quantity. Matching within each condition did not meet the 2% tolerance, so the actual edit L_2 per condition is reported with the comparison.

Only the shared-direction edits lower the L18 response. The endpoint is the per-case reduction of the L18 response energy, $1 - \|\Delta_{\text{edited}}\|^2/\|\Delta_{\text{unedited}}\|^2$, averaged with equal task weights; positive values mean a smaller response. The half-dose edit (Shared-small) lowers it by $4.2/3.6\%$ (Table S21).

The command change is small relative to the perturbation's. The edit's footprint, the edited-minus-unedited RMS of the first five commands (translation part) on the same input, is 0.0006 (Camera) and 0.0006 (headlight), about 0.5% and 1.2% of the change the perturbation itself causes (unedited perturbed-minus-clean RMS 0.128 and 0.051). Rotation and the continuous gripper value are separate endpoints. In the 120-case interim analysis the formatted gripper output was identical in all 3,840 non-self pairs although the continuous gripper value differed in 3,837 (an interim finding, not recomputed on the full population). This edit acts only at the first generation step; restoration, which acts at every step, moves the commands far more (Sec. S12.5).

The edit does not measurably change task success. Success is reported as bounds over missing outcomes, because the prespecified rule withholds point estimates and ordinary intervals whenever a planned outcome is missing (Table S21). Within each condition, the known lost and gained successes from unedited to Shared-large are $2.4/1.4$, $2.4/3.1$ and $1.4/3.1\%$ of planned pairs (clean/Camera/headlight; unknown pairs 1.7–2.1%). Table S23 gives the nine prespecified contrasts: Shared-large minus Unedited (D), minus Clean-feature (Q), or minus the mean of the three random sets (R), under each visual condition.

Over the 111 missing planned outcomes, only Camera Q keeps its sign under every assignment: it is positive, favoring the shared directions over the clean-feature directions. The other eight bounds include zero. The Clean-feature edit's own bounds exclude zero in each condition (Table S21): it lowers success under clean input and Camera and raises it under headlight removal. For the unedited policy, Camera's success bounds are 91.0–93.1%, against 92.4–93.4% under clean input, whereas headlight removal costs 2.8–5.9 points.

Post hoc sampling intervals on the known outcomes. A post hoc bootstrap of the 5,937 known outcomes over the 16 held-out tasks (2,000 draws, task-equal means inside each draw) assesses sampling variation. Table S23 reports its point estimates, pointwise 95% intervals and Bonferroni-adjusted intervals at $1 - 0.05/9$ beside the bounds. Camera Q and R have positive pointwise and adjusted intervals; the other seven intervals include zero. For Camera Q, the adjusted interval $[+0.3, +13.2]$ and the bounds $[+2.8, +6.6]$ both favor the shared directions over the clean-feature directions. For Camera R, the adjusted interval is positive while the bounds include zero, so the comparison with random directions depends on the analysis of known outcomes. Shared-large against Unedited includes zero in all three conditions.

Missing outcomes. Acquisition resumed in the prespecified order after a pause; interim results had been observed, but case selection did not depend on outcomes. 281 cases completed all 21 runs, 6 completed partially (90 missing runs), and one was interrupted after creation; none remained unexecuted. In total, 5,937 of 6,048 main outcomes are known. All planned outcomes remain in the estimand.

Development check of the schedule. With edits only at the first generation step, the L18 difference at the last step retains 3.7–5.5% of the first-step difference (12 development inputs); editing every generation step changes the commands about ten times more, with a reduction of about 2% in the visual difference (Spatial task 7, Camera).

Table S21. **Edits along the shared directions lower the L18 response; matched controls do not; all command footprints are small.** Prespecified 288-case plan, task-equal means. Edit L_2: actual edit norm per condition; L18 red.: per-case reduction of the L18 response energy (%); footprint: edited-minus-unedited RMS of the first five commands (translation) as a percentage of the unedited perturbed-minus-clean RMS; SR: closed-loop success over all planned episodes; ΔSR: edited minus unedited. SR and ΔSR are bounds (lower–upper, %) over every assignment of the 111 missing planned outcomes, so a range that excludes zero keeps its sign under every assignment; sampling intervals: Table S23. Random rows give the range over the three random direction sets.

Condition	Configuration	Edit L_2	L18 red. (%)	Footprint (%)	SR	ΔSR [bounds]
clean	Unedited	–	–	–	92.4–93.4	–
	Shared-large	148.4	–	–	90.6–92.4	$[-2.8, 0.0]$
	Shared-small	74.2	–	–	92.4–93.4	$[-1.0, +1.0]$
	Clean-feature	141.0	–	–	89.9–91.7	$[-3.5, -0.7]$
	Random-1/2/3 (range)	139.9–142.2	–	–	89.9–93.8	see text
Camera	Unedited	–	–	–	91.0–93.1	–
	Shared-large	132.8	8.8	0.5	91.7–93.8	$[-1.4, +2.8]$
	Shared-small	66.4	4.2	0.4	89.2–91.3	$[-3.8, +0.3]$
	Clean-feature	141.0	−2.5	0.6	87.2–88.9	$[-5.9, -2.1]$
	Random-1/2/3 (range)	139.7–141.9	0.2–0.3	0.5	89.2–93.1	see text
headlight	Unedited	–	–	–	87.5–89.6	–
	Shared-large	143.0	7.6	1.2	89.2–91.3	$[-0.3, +3.8]$
	Shared-small	71.5	3.6	1.1	89.2–91.3	$[-0.3, +3.8]$
	Clean-feature	141.0	−3.6	1.4	90.3–92.7	$[+0.7, +5.2]$
	Random-1/2/3 (range)	139.8–142.1	0.1–0.2	1.1	89.2–92.0	see text

Table S22. **Definition of the edit: training mean, directions, dose, schedule and response measurement.**

Quantity	Definition	Role
μ_H	Clean action-expert L9 post-block H at generation step 0, averaged over the 1,404 training-task records of the cross-observation study with that study's record weights (summing to one); the 16 held-out tasks are excluded. One 1024-vector per action position (10×1024), subtracted position by position. Its first five positions agree with the cross-observation clean fit center to 4.9×10^{-4} (declared tolerance 2.6×10^{-3}).	Training mean
B	16 directions for every edit: the cross-observation L9 shared directions (Shared-small/large), its L9 clean-feature directions (Clean-feature) and three random orthonormal sets from PCG64 seeds 20261014–20261016 (Random-1/2/3); all use the same μ_H.	Directions
λ	Shared-large 0.7521; Shared-small 0.3760 (half); Clean-feature 0.1721; Random-1/2/3 0.9427, 0.9826, 1.0000. Rule: on training first queries, each set's mean edit L_2 (equal weight per task, then per condition) equals a common budget of 141.1 that every set attains with $\lambda \leq 1$ (Random-3 at $\lambda = 1$); fixed before the evaluation outcomes were observed. Matching within each condition exceeded the 2% tolerance and remains unmatched.	Matched mean edit
Schedule	The edit is applied at the first generation step ($j = 0$) of every policy query, at all ten action positions; later steps and all other layers run unchanged.	Intervention time
L18 response	Action-expert L18 post-block H at the first generation step of the episode's first policy query (before the first environment step), all ten positions and 1024 channels. Per case and condition, ΔH is the condition minus clean under the same configuration, and the reduction is $1 - \sum(\Delta H_{\text{edited}})^2 / \sum(\Delta H_{\text{unedited}})^2$; means average noise draws within start state, start states within task and tasks equally.	Response endpoint

Table S23. **Success contrasts: what the edit costs against the unedited policy, and how it compares with the control edits.** Every row is Shared-large minus the named comparator, in percentage points. Bounds enumerate the 111 missing outcomes of the 6,048-episode plan. Points and intervals use the 5,937 known outcomes (post hoc whole-task bootstrap, 16 tasks, 2,000 draws). The 99.44% intervals are Bonferroni-adjusted across the nine contrasts. Bounds and sampling intervals describe different sources of uncertainty. Random mean averages the three random direction sets.

Condition	Comparator	Bounds over missing outcomes	Known-outcome point	95% interval	99.44% interval
Clean	Unedited (D)	$[-2.8, 0.0]$	-1.0	$[-3.0, +1.0]$	$[-3.8, +1.7]$
Clean	Clean-feature (Q)	$[-1.0, +2.4]$	$+0.7$	$[-1.9, +3.3]$	$[-3.0, +4.8]$
Clean	Random mean (R)	$[-2.2, +1.2]$	-0.8	$[-2.5, +0.8]$	$[-3.5, +1.3]$
Camera	Unedited (D)	$[-1.4, +2.8]$	$+0.7$	$[-0.7, +2.1]$	$[-1.2, +2.8]$
Camera	Clean-feature (Q)	$[+2.8, +6.6]$	$+5.0$	$[+0.7, +10.2]$	$[+0.3, +13.2]$
Camera	Random mean (R)	$[-0.6, +3.2]$	$+1.6$	$[+0.6, +2.7]$	$[+0.3, +3.2]$
Headlight	Unedited (D)	$[-0.3, +3.8]$	$+1.4$	$[-1.6, +4.0]$	$[-2.8, +5.3]$
Headlight	Clean-feature (Q)	$[-3.5, +1.0]$	-1.4	$[-5.9, +1.4]$	$[-8.0, +2.4]$
Headlight	Random mean (R)	$[-2.5, +1.6]$	-0.7	$[-2.5, +0.9]$	$[-3.2, +1.4]$

S12. Restoration of the Shared L9 Directions

Does undoing the perturbation along the shared directions move the commands back toward the clean run's? Restoration tests this at L9, writing clean values along these directions into the perturbed run at every generation step. It needs a clean observation of the same physical state, so it is a controlled test, not a deployable method. On held-out tasks, command error falls against the unedited run and both controls, and the shared directions reach most of what replacing the whole L9 state does. Task success is higher by 1.85 and 3.33 points (intervals in Table S24); replacing the whole L9 state raises it further (post hoc, pointwise). This supports main Sec. 4.2, Table 2 and Figs. 1b and 4.

Six runs per case answer three questions: does restoration help at all (against the unedited run), must the directions be the shared ones (against clean-feature and random controls), and how much can this layer do (replacing the whole L9 state)? Fixed in advance were the site, task success as the primary endpoint, command error as the second, and six simultaneous comparisons per endpoint. The controls write less; with edit size matched in development, restoration is better than clean-feature directions under Camera but not under Sensor Noise (Sec. S12.5).

S12.1. Directions, tasks and cases

Directions and site. The edit acts on the action expert's L9 post-block state (ten ordered action positions, 1,024 channels). L9, the middle recorded layer, was fixed before the confirmation runs as a compromise between the response coverage of L1 and the command prediction of L18 (main Sec. 4.1); one site, and two of the four families (Camera and Sensor Noise), keep six runs per case tractable. The shared and clean-feature directions are the first sixteen directions of the position-centered LIBERO-Plus fits of Sec. S3 (each plan position's mean removed before fitting), learned at the first query and first generation step and shared by all positions.

Held-out tasks. Development uses folds 0 and 3, confirmation folds 1, 2 and 4: nine and 31 tasks with no overlap, so that choices made in development are tested on unseen tasks. Each task uses the fit from its fold of the original split, which held out a task component (tasks sharing a clean scene) with a family; neither is in that fit's training data. Camera confirmation has 27 tasks and 48 settings; Sensor Noise has 30 tasks and 60 settings; the two share 26 tasks. Each setting has one start state, two policy-noise draws and six runs, giving $108 \times 2 \times 6 = 1{,}296$ episodes. All planned command measurements and outcomes are recorded.

S12.2. Restoration and the six runs

A clean observation of each run's own current state. Runs move apart after the first query, so each needs a clean

observation of its own state. Let s_q^a be the physical and controller state of run a at query q, and x_{qj}^a, t_j its current plan and time at generation step j. Both observations are rendered from this s_q^a. For Camera, the clean observation restores the setting's complete clean visual configuration; for Sensor Noise, it removes the image transform from the same underlying image. Rendering it leaves the simulator, controller and random-number states unchanged.

Two forward passes with these observations and the same x_{qj}^a, t_j give $H_T, H_C \in \mathbb{R}^{10\times1024}$ at L9, so H_C is the clean value for the run's current state. Restoration sets

$$D = H_T - H_C, \qquad H' = H_T - \alpha DBB^\top, \qquad \alpha = 1, \tag{S17}$$

where $B \in \mathbb{R}^{1024\times16}$ has orthonormal columns. Up to the precision of the write, $H'B = H_CB$, and the orthogonal complement stays perturbed. Position centering is used only to fit B; the edit projects the full current difference D onto the fixed directions. After the write, later blocks continue with the perturbed image prefix.

The six runs. All six runs start from the same simulator and controller state and use the same policy noise at each query. The clean run N_C (the target) and the perturbed run N_T are unedited. Restoration R_{resp} and the controls R_{clean} and R_{rand} apply Eq. (S17) with the shared, clean-feature and random directions, at all ten generation steps of every policy query, with $\alpha = 1$. R_{full} replaces the whole L9 state, $H' = H_C$, at the same site and schedule; its later blocks also read the perturbed prefix. Once an edit changes generation, each run follows its own plan and trajectory.

The random control uses a fixed mixture of three random sets of 16 orthonormal directions, assigned within task and family, with both noise draws of a setting receiving the same set. One set is the first sixteen columns of the 32-direction random set of Sec. S3; the other two come from the same generator with separately fixed seeds.

S12.3. Endpoints and simultaneous intervals

Command error. Let $A_{tvn}^a \in \mathbb{R}^{5\times3}$ hold the translation parts of the first five commands of the first query, for task t, setting v, noise draw n and run a. The command error is

$$e_{tvn}^a = \left\| A_{tvn}^a - A_{tvn}^{N_C} \right\|_F, \tag{S18}$$

the Euclidean norm over fifteen numbers, in command units after the adapter. All 1,296 records contain the five commands, logged after the environment step and identical to the first five positions of the ten-position plans. The clean target is paired by setting, start state and noise draw.

Task-equal contrasts and fixed scales. Positive contrasts favor restoration. Write $\langle z \rangle_f$ for the average over the two noise draws, then over settings within a task, then equally over the tasks of family f. For comparator $b \in \{N_T, R_{\text{clean}}, R_{\text{rand}}\}$, the command-error reduction and the success change are

$$G_{f,b} = \frac{\langle e^b - e^{R_{\text{resp}}} \rangle_f}{S_f}, \tag{S19}$$

$$\Delta_{f,b}^Y = 100 \left\langle Y^{R_{\text{resp}}} - Y^b \right\rangle_f, \tag{S20}$$

where Y is the final binary outcome. The family scales, fixed in advance, are $S_{\text{Camera}} = 0.316943$ and $S_{\text{Sensor}} = 0.200099$ in the same command units. They are constants, so G is in units of S_f, not a percentage of each case's error. Main Figs. 1b (against N_T) and 4a (against all three) show these contrasts in command units, with each gain and interval endpoint multiplied by S_f.

Statistics: six simultaneous comparisons for each endpoint. Each endpoint has its own family of six comparisons: three comparators under two perturbation families. The prespecified procedure uses 10,000 bootstrap draws of whole tasks, keeping all paired observations of each drawn task. Within a family, the same task draws serve all comparators and both endpoints. The simultaneous 95% intervals use percentile tails $0.05/(2\times6)$ and $1 - 0.05/(2\times6)$, with linear interpolation between ordered draws.

S12.4. Results: command error and task success

Command error falls against the unedited run and both controls. All six command intervals are above zero (Table S24). Against N_T, restoration lowers the task-equal command error from 0.4093 to 0.3120 under Camera and from 0.2091 to 0.1987 under Sensor Noise. In units of S_f, it reduces the error by 0.2908 and 0.0337 more than the clean-feature control, and by 0.2999 and 0.0516 more than the random control.

The reduction is broad across tasks. Against N_T, the command error falls in 25 of 27 Camera tasks and 24 of 30 Sensor Noise tasks. The table below summarizes the task-level reductions behind Table S24, as comparator minus R_{resp} in command units (the units of main Figs. 1b and 4a), with tasks weighted equally; > 0: tasks where restoration has the lower error. Medians lie below the means: the three largest task reductions account for 37% (Camera) and 48% (Sensor Noise) of the summed reduction. These are descriptive; the simultaneous intervals are in Table S24.

	Camera (27 tasks)			Sensor Noise (30 tasks)		
Comparator	Mean	Median	> 0	Mean	Median	> 0
N_T	0.0972	0.0557	25	0.0105	0.0060	24
R_{clean}	0.0922	0.0538	23	0.0067	0.0041	23
R_{rand}	0.0951	0.0552	25	0.0103	0.0057	24

Table S24. **Restoration against the unedited run and both controls: command error falls in all six comparisons; task success is higher against the unedited run.** Positive values favor restoration: first five commands closer to those of the clean run N_C (in units of a fixed family scale S_f), or more successes. Each endpoint has its own family of six simultaneous 95% whole-task bootstrap intervals. Camera: 27 tasks, 48 settings and 96 cases; Sensor Noise: 30 tasks, 60 settings and 120 cases. Every success interval includes zero; against R_{clean} under Sensor Noise its lower end is zero.

Family	Comparator	Command-error reduction $G_{f,b}$ [simultaneous interval]	Success change (pp) [simultaneous interval]
Camera	N_{T}	0.3068 [0.1573, 0.4821]	+1.85 [−8.33, 11.11]
Camera	R_{clean}	0.2908 [0.1423, 0.4641]	−1.85 [−12.04, 7.41]
Camera	R_{rand}	0.2999 [0.1530, 0.4724]	+1.85 [−9.26, 12.96]
Sensor Noise	N_{T}	0.0523 [0.0179, 0.0985]	+3.33 [−1.67, 9.17]
Sensor Noise	R_{clean}	0.0337 [0.0035, 0.0740]	+3.33 [0.00, 8.33]
Sensor Noise	R_{rand}	0.0516 [0.0165, 0.0996]	+1.67 [−2.50, 6.67]

Under Camera, restoration moves the commands mainly toward the clean commands. To separate progress toward the clean commands from other change, we split each run's change in two. Write the translation parts of the first five commands as $A^a \in \mathbb{R}^{15}$. For each setting, start state and noise draw, let $c = A^{N_C} - A^{N_T}$ be the direction toward the clean commands and $d^a = A^a - A^{N_T}$ the change the edit causes. Its signed component along c and the size of the rest are

$$d^a_{\parallel} = \frac{\langle d^a, c\rangle}{\|c\|_2}, \qquad d^a_{\perp} = \left\| d^a - d^a_{\parallel} \frac{c}{\|c\|_2} \right\|_2 . \tag{S21}$$

For each run, the command error satisfies $(e^a)^2 = (\|c\|_2 - d^a_{\parallel})^2 + (d^a_{\perp})^2$: $d_{\parallel}$ is progress toward the clean commands and $d_{\perp}$ the change in other directions. Both are computed per record and then averaged like the command error.

All 216 clean-target distances are positive. Table S25 reports all six runs as a descriptive decomposition. Restoration has mean along/orthogonal components $0.10365/0.04660$ under Camera, so most of its change points toward the clean commands, and $0.01139/0.01229$ under Sensor Noise, where the change in other directions is as large as the progress toward the clean commands. Only the command-error contrasts carry intervals (Table S24).

The 16 directions reach most of what replacing the whole L9 state does. Replacing the whole L9 state lowers the mean command error to 0.2973 (Camera) and 0.1943 (Sensor Noise), only somewhat below restoration (Table S26); the remaining error enters through the later blocks, which still read the perturbed image. These values are a reference, not one of the six comparisons. From Tables S26 and S25, replacing the whole state removes 27% (Camera) and 7% (Sensor Noise) of the command error of N_T, restoration 24% and 5%, so restoration reaches 86.9% and 70.7% of the whole-state reduction (from unrounded values, as in the main paper).

Edit size: the controls write less. The controls share restoration's number of directions, coefficient and schedule, but each copies only the clean–perturbed difference along its own directions; the shared directions, fitted to such differences, write five to seven times as much (Table S26). Because a larger edit might by itself correct more, development compares the sets at matched edit size (Sec. S12.5), where restoration is better than clean-feature directions under Camera but worse under Sensor Noise. Restoration's edit size is 62% and 71% of the whole-state edit, so per unit of edit it removes 1.4 times as much command error as replacing the whole state under Camera and as much under Sensor Noise.

Task success is also higher. Observed success is higher with restoration: 75.00% against 73.15% under Camera and 86.67% against 83.33% under Sensor Noise. The clean runs succeed in 98.15% and 97.50% (Table S26). The paired gains over N_T are +1.85 and +3.33 percentage points. All six simultaneous success intervals include zero, the one against the clean-feature control under Sensor Noise with its lower end at zero (Table S24). Under Camera, the clean-feature control has higher observed success than restoration.

Replacing the whole L9 state raises success. In a post hoc analysis with pointwise intervals, outside the six simultaneous comparisons, R_{full} minus N_T is +7.41 [+0.93, +14.81] (Camera) and +5.83 [0.00, +12.50] (Sensor Noise): this layer matters for the outcome, and restoration delivers part of that gain.

Lost and gained successes in every run. The table counts, unweighted, the cases whose outcome differs from the unedited run N_T with the same noise draw: failures that become successes (gained) and the reverse (lost). Gained successes grow with how much of the clean state is copied, while every edited run also loses some. The counts are descriptive; the simultaneous intervals are in Table S24.

Table S25. **How each run changes the first five commands: progress toward the clean commands and change in other directions.** Descriptive task-equal means on all confirmation cases: 96 Camera cases (48 settings, 27 tasks) and 120 Sensor Noise cases (60 settings, 30 tasks), two noise draws per setting. Each quantity is computed per record from the translation parts of the first five commands, then averaged. $d_{\parallel}, d_{\perp}$ are defined in Eq. S21; e is the command error. Command units; means without intervals. The geometric identity holds per record, not for the means.

	Camera			Sensor Noise		
Run	Along $d_{\parallel}$	Orthogonal $d_{\perp}$	Error e	Along $d_{\parallel}$	Orthogonal $d_{\perp}$	Error e
Clean (N_C)	0.40928	0.00000	0.00000	0.20913	0.00000	0.00000
Perturbed, unedited (N_T)	0.00000	0.00000	0.40928	0.00000	0.00000	0.20913
Restoration (R_{resp})	0.10365	0.04660	0.31205	0.01139	0.01229	0.19867
Clean-feature directions (R_{clean})	0.00529	0.01001	0.40422	0.00388	0.00633	0.20541
Random directions (R_{rand})	0.00221	0.00391	0.40710	0.00016	0.00246	0.20900
Whole L9 state (R_{full})	0.11974	0.05176	0.29733	0.01671	0.01918	0.19433

Table S26. **The six runs: command error, task success and edit size.** Descriptive task-equal means on the confirmation cases. e: command error (Eq. S18), zero for N_C by definition. Edit size: actual change written into the L9 state (tokenwise ℓ_2), summed over every query of the run's own episode, so it depends on both the writes and the episode length. R_{full} (whole L9 state) is a reference, not one of the six prespecified comparisons.

	Camera			Sensor Noise		
Run	Error e	Success (%)	Edit size (10^3)	Error e	Success (%)	Edit size (10^3)
N_{C}	0.0000	98.15	0.00	0.0000	97.50	0.00
N_{T}	0.4093	73.15	0.00	0.2091	83.33	0.00
R_{resp}	0.3120	75.00	46.20	0.1987	86.67	51.47
R_{clean}	0.4042	76.85	7.79	0.2054	83.33	7.74
R_{rand}	0.4071	73.15	9.65	0.2090	85.00	9.64
R_{full}	0.2973	80.56	74.20	0.1943	89.17	72.77

	Camera (96 cases)			Sensor Noise (120 cases)		
Run	Successes	Gained	Lost	Successes	Gained	Lost
N_T	69	–	–	100	–	–
R_{rand}	68	3	4	102	5	3
R_{clean}	72	6	3	100	3	3
R_{resp}	71	7	5	104	7	3
R_{full}	76	10	3	107	10	3
N_C	94	26	1	117	19	2

S12.5. Development checks: generation schedule and matched edit size

On the nine development tasks, restoring at every generation step matters. With edit size matched, restoration beats clean-feature directions under Camera but not under Sensor Noise; the confirmation runs use the unmatched edit ($\alpha = 1$). These are descriptive means without intervals.

Saved states and plan endpoint. Camera covers eight settings on five tasks; Sensor Noise covers seventeen settings on nine tasks. Each setting has two noise draws and four saved query states, $q = 0, 2, 5, 10$, from the unedited perturbed run, giving 64 Camera and 136 Sensor Noise cases, all complete. Clean and perturbed observations share the saved physical state, every candidate generation starts from the same recorded policy noise, and each case keeps its assigned directions.

The endpoint is the first five positions of the generated translation plan, and its improvement is

$$g^a = \frac{\|A^{N_T} - A^{N_C}\|_F - \|A^a - A^{N_C}\|_F}{S_f}, \qquad \text{(S22)}$$

with the fixed family scales defined above. Means average the four saved queries and both noise draws within settings, then settings within tasks, with equal weight per task (Table S27). This endpoint is a plan at a saved state; the confirmation endpoint above is the first five commands of a complete rollout.

Restoring at every generation step matters. Both schedules use Eq. (S17) with the shared directions. Moving from restoring only at $j = 0$ to restoring at $j = 0, \ldots, 9$ raises the mean gain from 0.00475 to 0.15468 for Camera and from 0.00188 to 0.06687 for Sensor Noise. The schedule changes both the sequence of writes and the later generation states. The edit on original LIBERO, which acts only at the first step, is reported in Sec. S11.

Table S27. **Development: plan gains depend on the schedule and on the edit-size rule.** Descriptive task-equal means of Eq. S22 on all 64 Camera and 136 Sensor Noise cases: 8/17 settings on 5/9 tasks, two noise draws and saved queries $q = 0, 2, 5, 10$. First/all: edit at $j = 0$ or at $j = 0, \ldots, 9$. Matched: Eq. S23 applied at each run's current state. N_T has zero gain by definition.

Rule	Camera / S_f	Sensor Noise / S_f
Clean (N_C)	1.20070	1.19909
Unedited (N_T)	0.00000	0.00000
Shared, first	0.00475	0.00188
Shared, all	0.15468	0.06687
Clean-feature, all	0.02056	0.02137
Random, all	0.00364	0.00118
Whole L9 state, all	0.22249	0.12432
Matched: shared	0.02539	0.00975
Matched: clean-feature	0.01763	0.01881
Matched: random	0.00308	0.00044

A common edit size at each position. The sets write equal amounts, so that they differ only in direction. For the shared, clean-feature and assigned random directions B_k, let $p_k = (H_T - H_C)B_k B_k^\top$ at one action position. Then

$$s = \min_{k \in \{\text{resp,clean,rand}\}} \|p_k\|_2, \qquad H'_k = H_T - s\frac{p_k}{\|p_k\|_2}. \tag{S23}$$

A zero target leaves the state unchanged. The actual written changes, $H_{\text{after}} - H_{\text{before}}$, are recorded; their shared-minus-clean-feature norm differences take both signs. Under complete generation, each run reapplies this rule at its own current state.

Clean-feature edits change the local output more. Applying each candidate write at the same state of the unedited perturbed generation, we measure the change in the 32-dimensional normalized flow velocity, $\delta v_k = v_k - v_{N_T}$, separately for plan positions 0–4 and 5–9. Clean-feature edits give a larger $\|\delta v\|_2$ than shared-direction edits in every recorded mean (one per task, query, generation step and position group): $400/400$ for Camera ($5 \times 4 \times 10 \times 2$) and $720/720$ for Sensor Noise ($9 \times 4 \times 10 \times 2$). These grids repeat measurements of the same five and nine tasks. The clean local velocity was not recorded.

With edit size matched, the ranking differs between families. Under complete generation with matched edit size, the gains for shared/clean-feature directions are $0.02539/0.01763$ for Camera and $0.00975/0.01881$ for Sensor Noise. Restoration is thus better than clean-feature directions under Camera and worse under Sensor Noise, although clean-feature directions change the local output more in both: local output size is not plan correction.

Implementation notes. Projections and edit sizes are computed at higher precision than the policy's own execution, which writes the edited state at its native precision; writes that round to zero at that precision remain in the data.

S13. How the Readout Estimator Changes Command Comparisons

Does the finding that commands are read best at L18 depend on the linear readout? We refit it with two nonlinear estimators on the same coordinates, pairs and splits. The layer ordering holds: the random-feature readout of the shared coordinates has higher translation error than predicting no change in 4 of 4 families at L1, 1 of 4 at L9 and 0 of 4 at L18 (Table S28). Which directions predict better, however, depends on the estimator: of 216 comparisons, the random-feature readout changes the interval category (above, spanning or below zero) of 102 and the MLP of 76. This supports Sec. 4.1 of the main paper (Fig. 3b): a linear readout predicts commands best at L18 and poorly at L1 and L9, and rankings depend on the readout.

What was compared. Only the estimator differs from the main paper, which uses a fixed affine ridge readout (Sec. S3.4). Here a random Fourier feature (RFF) ridge readout and a one-hidden-layer MLP read the same 16 coordinates along the shared and the clean-feature directions, on the same 404 pairs and held-out splits. The 216 comparisons per readout are the 36 layer–family–command cells, each comparing the shared directions with the full hidden state, the magnitude baseline, the centered clean-feature directions, random directions and the uncentered shared directions, plus first flow (the clean–perturbed difference of the step-0 update) with the shared coordinates against first flow alone. Table S29 adds rotation and gripper.

The ranking of shared over clean-feature directions depends on the estimator. Under the random-feature readout the L9 advantage of the shared directions becomes a zero-spanning difference in Background, Camera and Sensor Noise ($+49.6\% \to -0.1\%$, $+27.4\% \to -3.7\%$, $+56.3\% \to -5.0\%$) and stays positive in Light ($+71.0\% \to +49.8\%$; Table S29). At L1 the affine advantage ($+20.3$ to $+74.7\%$, all four intervals above zero) stays positive under random features in Light ($+21.9\%$) and Sensor Noise ($+17.1\%$), and under the MLP in all four families. Conversely, for Camera translation at L18, the random-feature readout favors the shared directions by 13.4% $[7.1, 20.5]$, where the affine interval spans zero. Its error is 10.909×10^{-3}, against 3.737×10^{-3} for the affine readout and 15.414×10^{-3} for no change (Table S28). At L18 under the MLP, the shared directions have higher translation error than

the clean-feature directions in Background and Camera, and higher rotation error in every family.

Absolute accuracy limits what the nonlinear readouts show. The MLP has higher error than predicting no change in every layer–family cell for every feature, first flow included (point estimates). Its comparison therefore describes this estimator and cannot determine how much command information the coordinates hold. For this reason the main paper reports the fixed affine readout, and relies on the removal test (Sec. S11) and restoration (Sec. S12) to measure directly how edits along the directions change the computation and the commands.

Implementation notes: matched fitting and resampling. The nonlinear-readout analysis was specified before its results were inspected. It uses the affine readout's 404 matched pairs, fixed position-centered shared, clean-feature and random directions, 60 joint task/family holdout splits, 2,000 whole-task bootstrap draws, pair weights and scoring. Each input block was centered by its outer-training weighted mean and divided by its scalar RMS, using all outer-training rows before any inner split; flow and response blocks were standardized separately before concatenation. The random map, the inner-fold assignment and the MLP were each seeded deterministically from the readout, family, layer, fold and feature block.

Random-feature ridge. The map was $\phi(z) = \sqrt{2/D}\cos(z\Omega/\sigma + b)$, with standard-normal entries in Ω, phases uniform on $[0, 2\pi)$, and $D = 2{,}048$ except for the ten-dimensional magnitude baseline ($D = 40$). The bandwidth σ was the median off-diagonal pairwise Euclidean distance among standardized outer-training rows, using one if the median was zero. Five-fold cross-validation over outer-training task groups, keeping tasks that share a clean scene together, selected $\lambda \in \{0.001, 0.01, 0.1, 1, 10\}$ by mean validation error; exact ties kept the smaller value. The map was fixed before inner fitting; ridge then re-centered and scalar-scaled the mapped features within each inner training fold, with an unpenalized intercept.

MLP fitting and stopping. The MLP had one hidden layer of 256 ReLU units and a linear 390-dimensional output in raw/adapter units, trained by full-batch Adam with learning rate 10^{-3}, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$ for at most 500 updates. A seeded permutation reserved approximately 20% of outer-training task groups for validation (rounded to the nearest integer, at least one); the targets were centered using the remaining fitting rows and their normalized pair weights. Initial weights were Gaussian with standard deviations $\sqrt{2/d}$ and $\sqrt{1/256}$ for the input and output layers (d is the input width), with zero biases. Validation MSE was checked after each update; training stopped after 25 consecutive updates without an improvement greater than 10^{-12}, and predictions used the best validation weights. The archive keeps task-level scores and aggregate results, but not the bandwidth, selected penalty or stopping update of each individual fit.

S14. Response Geometry and Command Information in π_0

Do different perturbations also change the features of π_0 along shared directions? Yes: on held-out tasks and families, 16 shared directions retain 70–77% of the response energy at L9, 39–50% at L1 and 43–52% at L18, against 1–28% for clean-feature and about 1.5% for random directions, in every family (Table S30). This is what main Sec. 4.1 means by "the result holds in π_0". Two things qualify it. First, the command readout: the shared coordinates carry command information at L18 but predict commands no better than clean-feature coordinates. Second, the paired executions stored no internal states, so the recordings come from a rerun on different hardware, admitted only where its actions stayed close to the original ones (369 of the 404 pairs).

Reading the table. Each row of Table S30 is one layer and family; higher is better. Means weight the 28 task components (groups of tasks that share a clean scene) equally.

The shared directions carry command information at L18 but do not beat clean-feature directions. A linear readout predicts the change in the first five commands (translation) from the 16 coordinates. At L18 it beats the magnitude baseline (positionwise response magnitudes) and random coordinates, with intervals above zero in 4 of 4 families for each comparison. The corresponding counts are 0 of 4 at L1 and 0 of 4 at L9. Against the full hidden state or the centered clean-feature coordinates, no interval is above zero at any of the three layers (0 of 4 each). Adding the shared coordinates to first flow (the clean–perturbed difference of the step-0 update) gives no interval above zero.

Details: regenerating the recordings. The policy was rerun on the recorded first-query LIBERO-Plus inputs and policy noise of the paired executions, on different hardware. The π_0 plan has 50 actions, and its action expert prepends one state token; the recorded state is $[10, 3, 1, 51, 1024]$. Regenerated raw actions differ from the recorded actions by at most 0.006–0.008 in absolute value. Input reconstruction is deterministic, and two independent reruns of the same run on the same hardware agree exactly.

Table S28. **Along the shared directions, the affine and random-feature readouts beat predicting no change at L18 in every family; the MLP never does.** Translation error (MSE $\times 10^3$) of the first five commands, task-equal on the same 404 matched pairs across 28 tasks; lower is better. Every estimator uses the same position-centered 16 shared and clean-feature directions; No-change is common to all. RFF ridge: random Fourier feature ridge readout. Error reductions with pointwise 95% intervals, including rotation and gripper: Table S29.

		Affine		RFF ridge		MLP		
Layer	Family	Shared	Clean	Shared	Clean	Shared	Clean	No-change
L1	Background	4.297	10.213	2.450	2.723	16.543	129.003	2.084
L1	Camera	16.280	20.428	16.404	17.002	21.546	100.024	15.414
L1	Light	1.875	7.400	1.303	1.668	4.418	76.428	1.063
L1	Sensor Noise	18.564	27.219	4.914	5.930	139.285	366.620	4.609
L9	Background	3.825	7.597	1.929	1.927	16.318	58.358	2.084
L9	Camera	14.407	19.831	18.427	17.776	89.104	104.278	15.414
L9	Light	1.365	4.713	0.968	1.930	4.285	25.489	1.063
L9	Sensor Noise	6.331	14.484	4.028	3.836	26.858	132.264	4.609
L18	Background	1.035	0.964	0.786	0.787	33.596	22.576	2.084
L18	Camera	3.737	3.457	10.909	12.590	155.875	90.506	15.414
L18	Light	0.433	0.496	0.321	0.307	11.097	9.487	1.063
L18	Sensor Noise	1.538	1.480	1.953	2.056	67.980	52.938	4.609

Details: fitting the directions and the command readouts. We use the joint task/family split and affine readout of Sec. S3.4. For each held-out family, five task-component folds exclude both that family and the test fold from fitting; testing uses their intersection. Common tasks keep the $\pi_{0.5}$ fold assignments. Each split fits position means and shared and clean-feature directions anew on its admitted π_0 fitting recordings. Fitting weights tasks equally, then families, settings, noise draws and positions within their parents.

At the first generation step, removing the state token leaves 50 ordered positions with 1,024 channels. Each set of 16 directions shares its channel basis across positions, giving 800 coefficients per layer; positionwise magnitudes, full hidden responses and first-flow differences use 50, 51,200 and 1,600 features. First flow plus the shared coordinates uses 2,400 features. Each block is centered on its weighted fitting mean and divided by one fitting RMS scale; affine ridge uses an unpenalized intercept and fixed $\lambda = 0.01$. The combined readout scales its two blocks separately. Layers are fitted separately, with position coordinates kept in each feature vector. We score the regenerated first five commands (after the output adapter) separately for translation, rotation and continuous gripper. Statistics: the intervals in Table S30 use 2,000 whole-task bootstrap draws.

Admission rule for rerun recordings. A rerun was admitted if its actions stayed within twice the largest difference observed when the original π_0 runs were repeated on the same input across machines: raw-action maximum absolute difference ≤ 0.018; first-five translation RMS $\leq 3.1 \times 10^{-3}$, rotation RMS $\leq 2.1 \times 10^{-3}$, gripper maximum $\leq 3.9 \times 10^{-3}$; sign agreement where the recorded value exceeds three times the threshold; no NaN. The rule depends on no outcome. It replaced an exact-match criterion after the discrepancy was observed.

Admitted pairs and exclusions. Of 808 runs, 769 were admitted and 39 excluded without retry: 7 runs of four Light settings were missing for technical reasons and 32 exceeded the tolerance (25 on the gripper threshold, 7 on the raw-action threshold, largest 0.27). This left 370 of the 404 planned clean/perturbed pairs on 28 tasks. One further pair was dropped, not retried, because the recorded state of an admitted run failed to write, so 369 pairs enter the analysis. Background, Camera, Light and Sensor Noise contribute 91/89/83/106 pairs on 28/28/27/28 task components. The exclusions spread over 21 family–task cells and all four families (Background 6 of 192 runs, Camera 15 of 204, Light 14 of 192 including the 7 technical, Sensor Noise 4 of 220). Of the 31 affected settings, 7 lost more than one run. Every statistic compares clean and perturbed recordings from the same rerun. The original actions serve only as the reference for admission, and admitted reruns lie within the original cross-machine scale rather than matching exactly.

S15. A Reusable Protocol for Responses and Paired Task Outcomes

How could another study run the paired evaluation of main Secs. 3.2 and 5, and what would it report? Each perturbed run is compared with the clean run that shares its policy noise, and the two clean runs give the clean rerun, the variation expected without any perturbation. Motion, lost and gained successes and the success change are reported against it, with bounds for missing outcomes. Section S15.1 recom-

Table S29. **Which directions predict commands better depends on the readout.** Error reduction (%) of the shared over the centered clean-feature directions for the first five commands, task-equal over 28 tasks, with pointwise 95% whole-task bootstrap intervals (2,000 draws shared across readouts); 404 pairs with matched commands across the four families (96, 102, 96 and 110 in Background, Camera, Light and Sensor Noise). RFF ridge: random Fourier feature ridge readout. Affine values agree with the main-paper readout at the displayed precision.

Layer	Family	Command	Affine	RFF ridge	MLP
L1	Background	translation	+57.9 [+45.4, +70.3]	+10.0 [−7.2, +29.9]	+87.2 [+83.6, +90.6]
L1	Background	rotation	+31.1 [+17.9, +45.2]	−0.3 [−12.8, +10.1]	+90.6 [+88.8, +92.4]
L1	Background	gripper	+57.7 [+40.4, +69.1]	+4.4 [−8.5, +13.8]	+87.4 [+83.1, +91.6]
L1	Camera	translation	+20.3 [+1.0, +33.7]	+3.5 [−3.6, +9.0]	+78.5 [+72.3, +83.7]
L1	Camera	rotation	+21.5 [+9.2, +32.6]	+5.9 [−2.3, +11.5]	+91.9 [+90.5, +93.2]
L1	Camera	gripper	+24.7 [+12.2, +34.3]	−7.5 [−17.5, −0.2]	+92.3 [+90.4, +93.9]
L1	Light	translation	+74.7 [+64.4, +82.5]	+21.9 [+3.1, +37.1]	+94.2 [+92.4, +95.7]
L1	Light	rotation	+68.4 [+58.5, +76.9]	+13.3 [+0.7, +25.4]	+96.1 [+95.6, +96.5]
L1	Light	gripper	+53.2 [+41.1, +62.3]	+6.4 [−5.5, +16.1]	+96.3 [+95.1, +97.2]
L1	Sensor Noise	translation	+31.8 [+8.6, +48.3]	+17.1 [+2.3, +29.6]	+62.0 [+54.0, +68.8]
L1	Sensor Noise	rotation	+39.5 [+14.8, +56.6]	+6.1 [−3.2, +19.7]	+68.3 [+58.9, +74.6]
L1	Sensor Noise	gripper	+31.7 [−0.9, +55.0]	−23.9 [−34.2, −12.4]	+66.2 [+49.6, +77.0]
L9	Background	translation	+49.6 [+25.6, +67.9]	−0.1 [−31.0, +19.1]	+72.0 [+60.7, +80.8]
L9	Background	rotation	+34.2 [+11.3, +53.8]	−1.9 [−13.2, +8.6]	+79.7 [+73.2, +85.1]
L9	Background	gripper	+30.8 [+17.9, +42.1]	+0.9 [−13.7, +14.6]	+81.6 [+71.7, +88.6]
L9	Camera	translation	+27.4 [+5.7, +40.3]	−3.7 [−9.8, +5.5]	+14.6 [−8.1, +30.9]
L9	Camera	rotation	+5.0 [−12.9, +17.9]	−8.7 [−16.9, +2.2]	+15.5 [−9.9, +35.0]
L9	Camera	gripper	+25.9 [+5.6, +39.9]	−4.7 [−20.7, +9.2]	+20.7 [−6.2, +43.9]
L9	Light	translation	+71.0 [+63.3, +77.5]	+49.8 [+30.1, +59.5]	+83.2 [+78.4, +86.9]
L9	Light	rotation	+54.0 [+38.2, +66.3]	+19.7 [+5.0, +34.0]	+82.5 [+77.8, +86.2]
L9	Light	gripper	+42.0 [+29.4, +52.1]	+6.7 [+0.2, +12.8]	+82.6 [+77.0, +86.8]
L9	Sensor Noise	translation	+56.3 [+33.1, +72.1]	−5.0 [−26.8, +9.4]	+79.7 [+75.9, +83.2]
L9	Sensor Noise	rotation	+59.2 [+50.6, +69.5]	−12.6 [−46.1, +2.6]	+80.0 [+77.0, +82.6]
L9	Sensor Noise	gripper	+51.5 [+40.7, +61.5]	+0.4 [−7.6, +7.1]	+84.9 [+78.8, +89.1]
L18	Background	translation	−7.4 [−38.2, +18.6]	+0.1 [−25.0, +23.5]	−48.8 [−103.0, −10.1]
L18	Background	rotation	+2.6 [−22.8, +25.1]	+22.3 [+4.6, +38.5]	−77.1 [−141.5, −22.3]
L18	Background	gripper	−33.5 [−69.1, −3.9]	−0.0 [−17.2, +14.4]	−71.8 [−156.0, −6.0]
L18	Camera	translation	−8.1 [−32.2, +15.5]	+13.4 [+7.1, +20.5]	−72.2 [−114.3, −38.5]
L18	Camera	rotation	−3.3 [−69.4, +34.9]	+21.2 [+12.6, +28.6]	−83.5 [−146.8, −35.7]
L18	Camera	gripper	−39.7 [−66.6, −16.0]	−19.3 [−40.1, −3.0]	−125.9 [−219.4, −58.6]
L18	Light	translation	+12.7 [−7.9, +31.0]	−4.4 [−46.1, +26.2]	−17.0 [−55.1, +13.1]
L18	Light	rotation	−13.3 [−71.3, +30.6]	+9.7 [−12.3, +29.0]	−36.5 [−75.5, −6.5]
L18	Light	gripper	−10.7 [−35.1, +5.9]	+4.0 [−10.0, +16.9]	−37.1 [−98.9, +5.1]
L18	Sensor Noise	translation	−3.9 [−51.5, +25.8]	+5.0 [−11.2, +16.9]	−28.4 [−67.2, +7.3]
L18	Sensor Noise	rotation	+21.2 [−18.6, +54.1]	+28.9 [+16.5, +45.5]	−28.1 [−53.5, −2.3]
L18	Sensor Noise	gripper	−18.7 [−39.6, +1.1]	+6.8 [−9.4, +20.1]	−75.0 [−155.6, −22.3]

putes our estimates from the recorded case-level data, which needs no policy (producing those data does), and lists our implementation checks.

1. Fix the plan before observing outcomes, so that nothing is chosen by its result. Declare the model and checkpoint, task and visual variant, matched clean rendering, instruction, initial state, two policy-noise draws (0 and 1), episode budget and source code. Run all runs of a setting on one machine, using the five runs below in this order. Record missing or invalid runs against the plan. Policy, execution and setting together identify a comparison.

2. Record plans, commands and motion separately. Record the policy's raw actions, the planned commands after the output adapter, the commands sent to the environment and each call's return. Differences in the first five commands (translation, rotation, gripper) are in adapter command units; their root mean square (RMS) gives the size of a change. Record the end-effector position (world meters) and orientation (radians, with the quaternion convention) after returned controls 0, 5, 15, 30 and 60. Record each run's final pose, its step index and whether the two runs of a pair have the same number of steps, and early stops, unreturned controls, missing poses and unknown outcomes Y explicitly. Fixed-step and final-pose comparisons use the recorded pose at that step. A returned command records what was sent to the controller; the signal it physically applied requires separate logging.

3. Compare the perturbation with the clean rerun. For

Table S30. **In π_0, the 16 shared directions retain far more of an unseen response than clean-feature or random directions.** Share of the response energy around the fitting-data mean (%) at the first query, task-equal mean on each family's measured pairs with pointwise 95% whole-task bootstrap intervals; 369 pairs on 28 task components overall (family counts in text); directions fitted without the test tasks and family. The recordings come from a rerun of π_0 on different hardware, admitted by the action-difference criterion in the text.

Layer	Family	Shared	Clean-feature	Random
L1	Background	44.4 [41.1, 48.0]	2.0 [1.8, 2.1]	1.4 [1.4, 1.5]
L1	Camera	49.2 [43.9, 54.8]	2.2 [2.1, 2.3]	1.5 [1.4, 1.6]
L1	Light	38.5 [35.9, 41.4]	2.2 [2.1, 2.2]	1.5 [1.5, 1.6]
L1	Sensor Noise	50.1 [46.2, 54.0]	2.0 [1.8, 2.2]	1.4 [1.3, 1.4]
L9	Background	72.2 [69.4, 75.1]	1.0 [0.9, 1.1]	1.8 [1.7, 1.9]
L9	Camera	77.2 [74.1, 80.0]	1.0 [0.9, 1.1]	1.7 [1.6, 1.8]
L9	Light	71.9 [69.6, 74.1]	1.1 [1.0, 1.2]	1.9 [1.8, 2.0]
L9	Sensor Noise	70.3 [66.3, 74.0]	1.7 [1.6, 1.8]	1.9 [1.8, 2.0]
L18	Background	43.7 [39.9, 48.1]	21.5 [18.8, 24.4]	1.5 [1.4, 1.6]
L18	Camera	51.9 [47.8, 55.5]	28.1 [24.0, 32.0]	1.5 [1.4, 1.6]
L18	Light	49.2 [46.2, 52.5]	17.5 [15.3, 20.0]	1.4 [1.3, 1.5]
L18	Sensor Noise	43.0 [39.2, 47.1]	21.5 [17.7, 25.9]	1.5 [1.4, 1.6]

Run	Visual input / noise	Purpose
R_0	matched clean / 0	clean run
T_0	perturbed / 0	perturbed, noise of R_0
R_1	matched clean / 1	clean rerun
T_1	perturbed / 1	perturbed, noise of R_1
R_0'	matched clean / 0	repeat of R_0

each noise draw, compare R_n with T_n. R_0–R_0' checks reproducibility. R_0–R_1 is the clean rerun, conditional on the two recorded draws, and T_0–T_1 is its perturbed counterpart. For valid binary outcomes, SF is a lost success ($Y_R = 1, Y_T = 0$) and FS a gained success. The success change is $\Pr(\mathrm{FS}) - \Pr(\mathrm{SF})$, and their sum is the rate of changed outcomes. Weight the two noise draws one half each, settings equally within a task, and tasks equally. Report bounds over missing outcomes and whole-task bootstrap intervals, adjusted for four families where stated, for sampling uncertainty.

4. State where each internal quantity is recorded, and keep recorded and recomputed states separate. For optional internal recordings of $\pi_{0.5}$, state for every scalar its layer, token region, generation step and recording path. Recomputing the prefix records the pre-block state H, the normalized state Z and the projected cached values V (fixed within a query). Recomputing the attention records, at the first query and generation step, the full-key probability P, the output after projection and before the gate, W, and the block-end action-expert state H. Keep L1, L9 and L18 and the main-image, wrist-image and text regions where available, and state whether a measure is raw or relative. Reconstruction coefficients describe an observed response; prediction and interventions need their own recorded inputs and operations. The π_0 paired executions stored no internal states; Sec. S14 regenerates first-query states for 369 pairs, and the protocol evidence for π_0 is commands, motion at fixed steps and outcomes Y.

5. Report task cost on its matched support. For every comparison with the clean run, perturbed or clean rerun, state the settings, metric and time step it covers, including missing and undefined measurements, and pair the two means on their common cases. Post hoc SF/SS groups describe lost and kept successes among observed outcomes. Later data collections keep their own identities beside the original results, including negative, null and incomplete comparisons.

S15.1. Recomputing estimates from recorded case data

From the 536 policy–setting records, the analysis recomputes task-equal outcome estimates, bounds over missing outcomes and end-effector position and orientation comparisons after 5 and 60 controls. It checks that records are unique and complete, with five outcome cells per case and valid units and measurement status. It keeps the original task order and resampling scheme. The primary population has 202 planned settings per policy; a sensitivity analysis uses all 268.

Outcome estimates keep the planned denominator. For each case, all possible assignments of the unknown outcomes are enumerated before each comparison is formed. Each reported bound holds for its comparison alone; different comparisons can reach their endpoints under different assignments. Motion estimates require both noise-matched comparisons and the clean rerun at the same metric and time step. We average over noise draws, settings and tasks in that order, then divide the two task-equal means; a zero clean-rerun mean leaves the ratio undefined. Statistics: the success-change intervals use four-family-adjusted 98.75%

coverage, and other outcome and motion intervals are pointwise 95%. Bounds over missing outcomes describe what the missing data allow, not sampling uncertainty.

Implementation and verification. Each entry names what was checked and what agreed, on the stated inputs rather than the original executions; removal-test and π_0 entries also record data-collection exceptions.

Recomputed hidden states and velocities. One real input is run through two identical branches at generation steps 0, 5 and 9. The recomputed hidden states agree with the recorded ones; the recomputed velocity differs from the policy's by at most 0.03125 in absolute value, with relative errors 0.00367–0.00457. With identical branches, an error relative to the effect is undefined. P and W were not recorded during the policy's own generation, so attention has no tolerance or pass rate (Table S1).

Case-level analysis. Run from its instructions in an empty directory, the outcome and motion analysis reproduces 16 outcome panels and 128 metric/time/reference panels and passes 1,448 checks against the original values. The package includes the data dictionary, configuration, support tables, expected outputs and edge-case tests.

Policies. The configurations are `pi05_libero` and `pi0_libero`. Checksums of the checkpoint, normalization and tokenizer files of both policies are recorded in the source package.

Reconstruction. The uncentered 16-direction reconstruction results are reproduced to within 2.88×10^{-14} in absolute task-score difference.

Affine readout. The implementation reproduces 324 recorded full-panel MSE and target-energy entries with the uncentered directions, with maximum absolute discrepancy zero.

Nonlinear readouts. The affine readout inside the nonlinear-readout code (Sec. S13) reproduces the expected affine results with absolute differences below 10^{-15} in task-equal MSE and 2×10^{-13} in unscaled relative error reduction, in all four families.

Corrected readout support. The reported readouts use the protocol's 404 pairs with matched commands. A first run had evaluated Light on all 110 common-task pairs instead of the specified 96 with matched commands, which moved its affine values by up to 0.039 in relative error reduction. That first run is kept as a record; its other three families are unchanged.

Later-observation summaries. The shared, clean-feature and random summaries in Table S18 use the original averaging over noise, initial state and task and its intervals. Applying this averaging to all 940 available L18 episodes reproduces all 480 task means and 30 condition means.

Aggregation. The aggregation estimates and correlation intervals of the ten-condition study (Sec. S9.1) are reproduced from the recorded case-level scalars.

Removal test: edit size. In the removal test (Sec. S11), edit sizes are matched on the training average over the three conditions. Per condition, the differences exceed the 2% tolerance.

Removal test: data collection. Data collection was paused at 229 of 288 cases and resumed in the fixed dispatch order. No case was selected by outcome. Two interim summaries (at 120 and 220 cases) had been seen before the decision to resume, as the decision record in the package states.

Removal test: execution exceptions. One case was interrupted after initialization; another was replaced under the prespecified technical-failure rule after a startup synchronization failure. Both are identified in the execution records.

Removal test: numerical checks. An independent calculation reproduced the response means exactly. The definitions in Table S22 are transcribed from the run configuration, not recomputed. The package holds the training mean, directions, random-direction seeds, edit sizes and the post hoc outcome bootstrap on available cases with per-draw values.

π_0 rerun. The records of the π_0 rerun on different hardware and of its admission rule are included (Sec. S14).

S16. Complete Comparison Panels

This section gives complete versions of the main figures and adds no analysis. Figure S7 combines what the main paper splits across Figs. 1b, 3 and 4. Its new panel (c) gives the gain of shared over clean-feature directions in predicting commands, $1 - \mathrm{MSE}_{\mathrm{shared}}/\mathrm{MSE}_{\mathrm{clean}}$; read it against the no-change reference in panel (b).

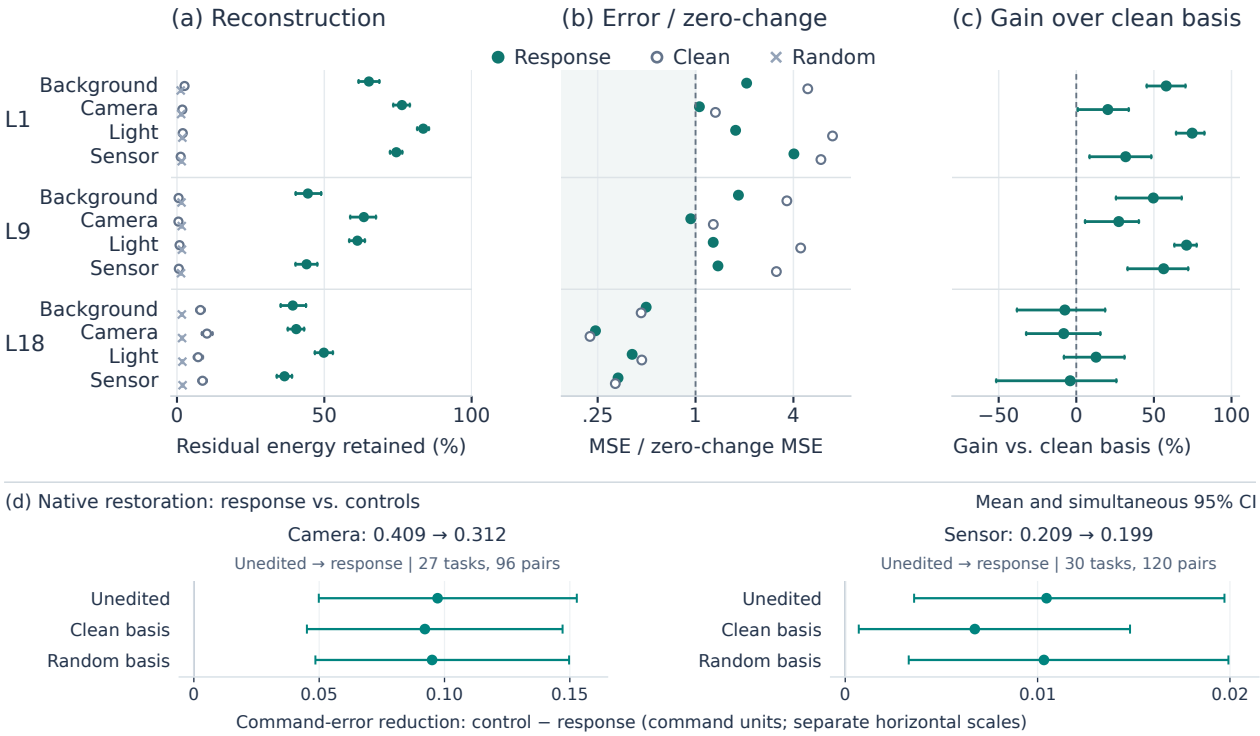


Figure S7. **Shared directions retain unseen responses, and restoring them at L9 reduces command error.** (a–c) The same 404 LIBERO-Plus pairs on 28 tasks; each fit leaves out the test tasks and family. (a) Share of the centered response energy at the first generation step retained by 16 shared directions. (b) Readout error relative to predicting no change; below 1 beats no change (log axis). (c) Gain over clean-feature directions in predicting commands; positive favors the shared directions. (a,c) Pointwise 95% task intervals. (d) Restoration at L9, a separate study: 96 Camera and 120 Sensor Noise cases on 27/30 tasks. Points show error reductions (other run minus restoration); intervals are simultaneous 95% over six comparisons. Headers give unedited → restoration error. Separate horizontal scales show the smaller Sensor Noise effect. All controls: Sec. S12.

Figure S8 shows every pair behind the rank correlations of main Sec. 5.1; compare the spread after five and 60 controls.

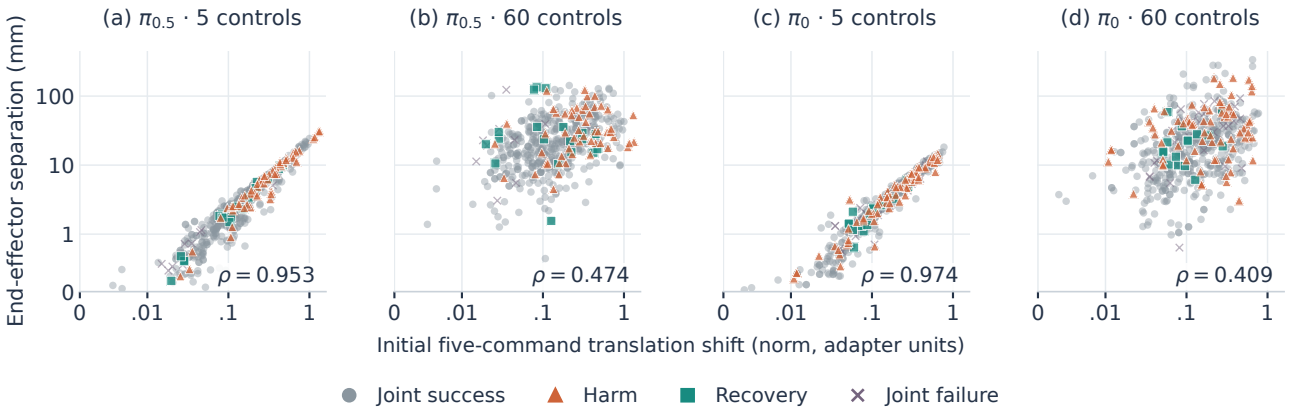


Figure S8. **The change in the first five commands tracks motion closely after five controls and more weakly after 60.** Paired executions: both times use the same 516/536 pairs for $\pi_{0.5}$ and 526/536 for π_0, on all 40 tasks. A pair enters if its commands and its motion at both times passed validation, without the cross-family restriction of Fig. 5. Color gives the observed paired outcome; all outcome classes are included. ρ: task-weighted rank correlation across the plotted pairs. Shared symmetric-log axes are linear below 0.01 command units and 1 mm. Pointwise 95% intervals and paired early-minus-late differences: Sec. S5.1.

Figure S9 gives all four outcome panels behind main Fig. 5. In panels (a) and (b), π_0 under Sensor Noise moves less than a clean rerun yet loses success.

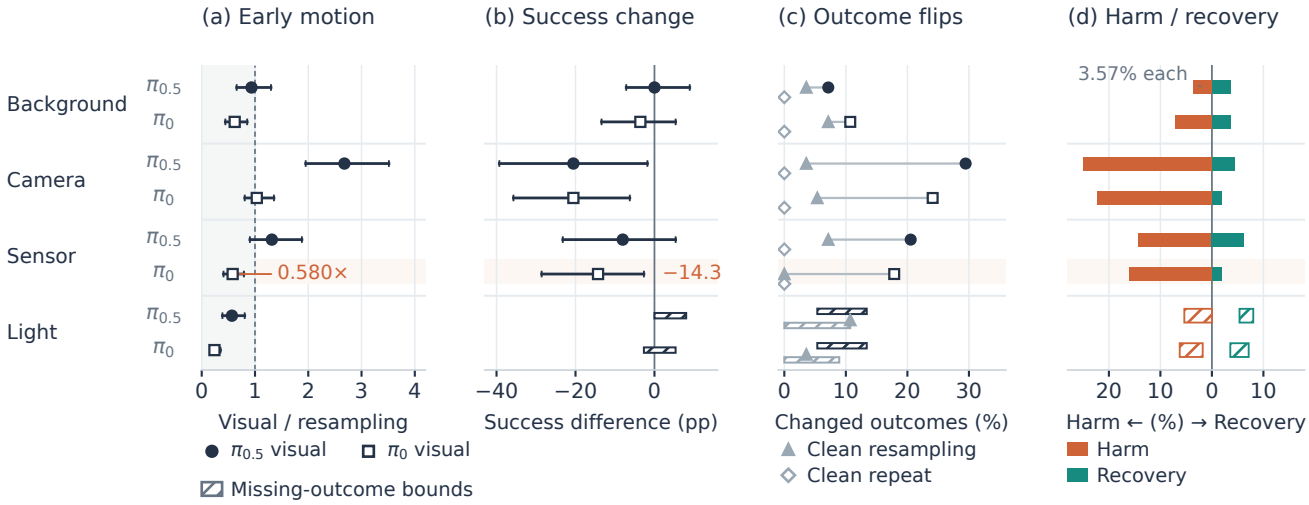


Figure S9. **Motion, net success and paired outcomes show different effects of a perturbation.** 202 settings with a LIBERO-Plus difficulty level, 28 tasks per policy. (a) Motion ratio: end-effector separation after five controls relative to that of the clean rerun. Ratios divide task-equal means, with pointwise 95% task intervals; shading marks values below the clean rerun. Light has 43/48 settings on 26 tasks ($\pi_{0.5}$), 44/48 on 27 (π_0). (b) Success change with four-family-adjusted 98.75% intervals. (c) Lines connect the rates of changed outcomes under the perturbation and under a clean rerun (descriptive). (d) Lost successes extend left and gained successes right; lengths are positive percentages. Hatching marks bounds over missing outcomes, distinct from sampling intervals. The highlighted row is π_0 Sensor Noise. Secs. S5 and S6.

S17. Sensitivity to the Aggregation of Response Energy

Does the C1 result depend on how response energy is averaged over tasks? No. When task energies are summed before taking the ratio, the shared directions retain 38.4–84.0% of the response energy, against 0.46–10.91% for clean-feature and 1.35–1.98% for random directions, and they exceed both controls in all 12 layer–family cells (Table S31). This supports main Sec. 4.1. It is a descriptive check with no new intervals; the task-equal score of the main paper remains the primary measure.

Two ways of aggregating. Let E_t be task t's mean squared response magnitude around the fitting-data mean at each plan position, and $S_{t,m}$ its mean squared reconstruction error for directions m. The main score gives tasks equal weight, $|T|^{-1}\sum_t(1 - S_{t,m}/E_t)$. The alternative sums errors and energies first,

$$R_m^{\text{energy}} = 1 - \frac{\sum_t S_{t,m}}{\sum_t E_t}, \tag{S24}$$

which gives more weight to tasks with larger responses. Both use the same saved predictions, fitting means and joint task/family holdouts, with nothing refitted. The inputs reproduce all 36 main layer–family–direction means and use the same tasks and energy denominators for all directions.

The result. The shared-direction values differ by at most a few percentage points between the two aggregations (Table S31). Both scores keep the fitting-data position mean, so their response energy includes offsets between fitting and held-out families.

		Main	Energy summed over tasks		
Layer	Family	Shared	Shared	Clean-feature	Random
L1	Background	65.14	66.39	2.27	1.43
L1	Camera	76.35	77.45	1.70	1.38
L1	Light	83.58	84.00	1.89	1.84
L1	Sensor Noise	74.44	76.98	1.10	1.60
L9	Background	44.42	42.62	0.52	1.51
L9	Camera	63.42	67.30	0.46	1.68
L9	Light	61.23	61.01	0.88	1.69
L9	Sensor Noise	43.98	47.22	0.64	1.35
L18	Background	39.29	38.40	8.73	1.66
L18	Camera	40.47	41.80	10.91	1.75
L18	Light	49.84	49.36	8.01	1.75
L18	Sensor Noise	36.50	38.90	8.93	1.98

Table S31. **The shared directions retain far more response energy than the controls under either aggregation.** Share of the response energy around the fitting-data mean (%) on the same 404 held-out pairs and 28 tasks. Main: the main paper's mean of task ratios. The last three columns take the ratio after summing task-level errors and energies. Each cell includes all 28 tasks, with 96/102/96/110 pairs for Background/Camera/Light/Sensor Noise. Descriptive values without intervals.

References